Academic Regulations

Course Structure & Detailed Syllabus

CHOICE BASED CREDIT SYSTEM

R22

COMPUTER SCIENCE
&
INFORMATION TECHNOLOGY

Bachelor of Technology (B.Tech)

For the batches admitted from the A.Y. 2022-23

I

MARRI LAXMAN REDDY
GROUP OF INSTITUTIONS

MLR Institute of Technology

(Autonomous)
Laxman Reddy Avenue, Dundigal
Hyderabad – 500043, Telangana State
www.mlrit.ac.in, Email: director@mlrinstitutions.ac.in

VISION OF THE DEPARTMENT

To emerge as a center of academic virtue with human and ethical values in the field of Computer Science and Information Technology to meet international standards.

MISSION OF THE DEPARTMENT

M1. To build the finest quality IT professionals by imparting value education, quality training, and hands on experience.

M2. To provide quality infrastructure through labs and other resources to continuously upgrade to the latest technology requirements.

M3. To promote and nurture the spirit of innovation and entrepreneurship in our students

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S):

PEO I: Graduates will acquire capability to apply their knowledge and skills to solve various kinds of computational engineering problems.

PEO II: Graduates will exhibit the ability to apply the acquired skills in various domains and multi-disciplinary areas, to function ethically and social challenges.

PEO III: Graduates will acquire soft skills to adapt and excel in diverse global environment.

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FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism,** independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

PRINCIPAL

CONTENTS

ACADEMIC REGULATIONS (R22)

B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2022-23)

1.0 For pursuing four year Under Graduate Degree Programme of study in Engineering & Technology (UGP in E&T) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2022-23 onwards. Any reference to "Institute" or "College" in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

2.0 Eligibility for Admission

Admission to the undergraduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government (EAMCET) or the College or on the basis of any other order of merit approved by the University/TSHE, subject to reservations as prescribed by the government from time to time.

The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA $\geq$ 5) required for the completion of the undergraduate programme and award of the B.Tech. degree.

UGC/ AICTE specified definitions/ descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms, which are listed below.

Semester Scheme

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks ($\geq$ 90 instructional days) each and in each semester - 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)' under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

Credit Courses

All subjects/ courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/ course in an L: T: P: C (lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/ week/ semester for Theory/ Lecture (L) courses or Tutorials.
- One credit for two hours/ week/ semester for Laboratory/ Practical (P) courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

Subject Course Classification

All subjects/ courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The College has followed almost all the guidelines issued by AICTE/UGC.

S. No.	Broad Course Classification	Course Group/ Category	Course Description
1	Foundation Courses (FnC)	BS–Basic Sciences	Includes Mathematics, Physics and Chemistry Subjects.
2		ES-Engineering Sciences	Includes Fundamental Engineering Subjects.
3		HS–Humanities and Social Sciences	Includes subjects related to Humanities, Social Sciences and Management.
4	Core Courses (CoC)	PC–Professional Core.	Includes core subjects related to the parent discipline/ department/ branch of Engineering.
5	Elective Courses (EℓC)	PE–Professional Electives	Includes elective subjects related to the parent discipline/ department/ branch of Engineering.
6		OE–Open Electives	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering.
7	Core Courses	Project Work	B.Tech. Project or UG Project or UG Major Project or Project Stage I & II.
8		Industry Training/ Internship / Industry Oriented Mini- project/ Mini- Project/ Skill Development Courses.	Industry Training/ Internship/ Industry Oriented Mini-Project/ Mini-Project/ Skill Development Courses.
9		Seminar	Seminar/ Colloquium based on core contents related to parent discipline/ department/ branch of Engineering.
10	Minor Courses	-	1 or 2 Credit Courses (subset of HS)
11	Mandatory Courses (MC)	-	Mandatory Courses (non-credit)

4.0 Course Registration

A 'faculty advisor or counselor' shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/ courses, based on their competence, progress, pre- requisites and interest.

The academic section of the college invites 'registration forms' from students before the beginning of the semester through 'on-line registration', ensuring 'date and time stamping'. The on-line registration requests for any 'current semester' shall be **completed before the commencement of SEEs (Semester End Examinations) of the 'preceding semester'**.

A student can apply for **on-line** registration, **only after** obtaining the '**written approval**' from faculty advisor/counselor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/ Counselor and the student.

A student may be permitted to register for all the subjects/ courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits (any 2 elective subjects), based on **progress** and SGPA/ CGPA, and completion ofthe '**pre-requisites'** as indicated for various subjects/ courses, in the department course structure and syllabus contents.

Choice for '**additional subjects/ courses**', not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.

If the student submits ambiguous choices or multiple options or erroneous entries during **on-line** registration for the subject(s) / course(s) under a given/ specified course group/ category as listed in the course structure, only the first mentioned subject/ course in that category will be taken into consideration.

Subject/ course options exercised through **on-line** registration are final and **cannot** be changed or inter-changed; further, alternate choices also will not be considered. However, if the subject/ course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternate choice either for a new subject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week** after the commencement of class-work for that semester.

Dropping of subjects/ courses may be permitted, only after obtaining prior approval from the faculty advisor/ counselor 'within a period of 15 days' from the beginning of the current semester.

Open Electives: The students have to choose three Open Electives (OE-I, II & III) from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered byhis own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.

Professional Electives: The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.

5.0 Subjects/ courses to be offered

A subject/ course may be offered to the students, **only if** a minimum of 15 students opt for it.

More than **one faculty member** may offer the **same subject** (lab/ practical may be included with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on - '**first come first serve** basis and CGPA criterion' (i.e. the first focus shall be on early **on-line entry** from the student for registration in that semester, and the second focus, if needed, will be on CGPA ofthe student).

If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide, whether or not to offer such a subject/ course for **two (or multiple) sections**.

In case of options coming from students of other departments/ branches/ disciplines (not considering **open electives**), first **priority** shall be given to the student of the '**parent department**'.

6.0 Attendance requirements:

A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/ courses (including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject.

Shortage of attendance in aggregate upto 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.

A stipulated fee shall be payable for condoning of shortage of attendance.

Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.

Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examinations of that semester. They get detained and their registration for that semester shall stand cancelled, including all academic credentials (internal marks etc.) of that semester**. They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking re-admission into that semester as and when offered; if there are any professional electives and/ or open electives, the same may also be re- registered if offered. However, if those electives are not offered in later semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.

A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in Item No. **6**.

A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course, if student secures not less than 35% (21 marks out of 60 marks) in the semester end examination, and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE(Semester End Examination) taken together; in terms of letter grades, this implies securing **'C'** grade or above in that subject/ course.

A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student isdeemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such 'one reappearance' evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

Promotion Rules

S. No.	Promotion	Conditions to be fulfilled
1	**First year first semester to first year second semester**	**Regular course of study of first year first semester.**
2	**First year second semester to Second year first semester**	**(i) Regular course of study of first year second semester.** **(ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.**
3.	**Second year first semester to Second year second semester**	**Regular course of study of second year first semester.**
4	**Second year second semester to Third year first semester**	**(i) Regular course of study of second year second semester.** **(ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.**
5	**Third year first semester to Third year second semester**	**Regular course of study of third year first semester.**
6	**Third year second semester to Fourth year first semester**	**(i) Regular course of study of third year second semester.** **(ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.**
7	**Fourth year first semester to Fourth year second semester**	**Regular course of study of fourth year first semester.**

A student (i) shall register for all courses/subjects covering 160 credits as specified andlisted in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing SGPA $\geq$ 5.0 (in each semester), andCGPA $\geq$ 5 (at the end of 8 semesters), (iv) **passes all the mandatory courses,** to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA **(at the end of undergraduate programme),** and shall be indicated in the grade card / marks memo of IV-year II semester.

If a student registers for '**extra subjects'** (in the parent department or other departments/branches of Engg.) other than those listed subjects totaling to 160 credits as specified in the course structure of his department, the performances in those '**extra subjects**' (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such '**extra subjects'** registered, percentage of marks and letter grade alone will beindicated in the grade card / marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items6 and 7.1 – 7.4 above.

A student eligible to appear in the semester end examination for any subject/ course, but absent from it or failed (thereby failing to secure **'C'** grade or above) may reappearfor that subject/ course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/ course will be carriedover, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in that subject.

A student **detained in a semester due to shortage of attendance may be re-admittedin the same semester in the next academic year for fulfillment of academic requirements**. The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/ CGPA calculations will be done for the entire semester in which the student has been detained.

A student detained **due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits.** The academic regulations under which the student has been readmitted shall be applicable to him.

8.0 Evaluation - Distribution and Weightage of Marks

The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).

In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours as follows:

1. Mid Term Examination for 30 marks:
   a. Part - A : Objective/quiz paper for 10 marks.
   b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks.

The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the concerned subject for 5 marks.

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination. The average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40%of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores ≥ 35% (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.

There is NO Computer Based Test (CBT) for R22 regulations.

The details of the end semester question paper pattern are as follows:

- The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an "either" "or" choice, which means that there will be two questions from each unit and the student should answer either of the two questions.
- The duration of Semester End Examination is 3 hours.
- For the subject, Computer Aided Engineering Graphics, the Continuous Internal Evaluation(CIE) and Semester End Examinations(SEE)evaluation pattern is same as for other theory subjects.

For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation:

1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
2. **10 marks for viva-voce (**or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.
4. The remaining 10 marks are for Laboratory Project, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other colleges which will be decided by the principal of the College.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
2. 15 for experiment/program
3. 15 for evaluation of results
4. 10 marks for presentation on another experiment/program in the same laboratory course and
5. 10 marks for viva-voce on concerned laboratory course.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40%of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores ≥ 35% (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.

The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I,II Semester course (*ex., Elements of CE/ME/EEE/ECE/CSE/Seminar & etc*): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course. Seminar evaluation conducted through day to day evaluation of the students in presentation/abstract.
2. For CSE/IT and allied branches the Continuous Internal Evaluation(CIE) will be for

marks. Each Mid-Term examination consists of two parts i) Part-A for 20 marks, ii)Part-B for 20 marks with a total duration of 2 hours.

Part-A:Objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 20 marks . **Part-B** : Descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks.

The remaining 10 marks of Continuous Internal Evaluation are for Assignment(5 Marks) and Subject Viva-Voce/PPT/Poster Presentation/Case Study (5 marks) and the evaluation pattern will remain same as for other theory subjects.

For all other branches, the Continuous Internal Evaluation (CIE) will be for 50 marks, Out of the 50 marks . Out of the 50 marks for Internal Evaluation:

a) A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks

b) 10 marks for viva-voce (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.

c) Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 15 marks.

d) The remaining 15 marks are for Laboratory Report/Project and Presentation, which consists of the Design (or) Software / Hardware Model Presentation(or) App completion of laboratory course and before semester end practical examination.

3. II Year II Semester *Real-Time (or) Field-based Research Project* course: The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (ii) secures less than 40% marks in this course.

There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.

The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his project work.

UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage – II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEE Theory examinations.

For Project Stage – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and

a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/ Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of project, Principal/COE selects an external examiner from the list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

A student shall be given only one time chance to re-register for a maximum of two subjects in a semester:

- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva- voce/PPT/ Poster presentation/ Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the class work in next academic year.

In the event of the student taking this chance, his Continuous Internal Evaluation marks for 40 and Semester End Examination marks for 60 obtained in the previous attempt stand cancelled.

For mandatory non-credit Audit courses, The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

For Mandatory Non-Credit Courses offered in a Semester, after securing ≥ 65% attendance and has secured not less than 40% of marks in the sum Total. Then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses '**Satisfactory'** or "**Unsatisfactory'** shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

9.0 Grading Procedure

Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/ Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.

As a measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

% of Marks Secured in a Subject/Course (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
Greater than or equal to 90%	O (Outstanding)	10
80 and less than 90%	A^{+} (Excellent)	9
70 and less than 80%	A (Very Good)	8
60 and less than 70%	B^{+} (Good)	7
50 and less than 60%	B (Average)	6
40 and less than 50%	C (Pass)	5
Below 40%	F (FAIL)	0
Absent	Ab	0

A student who has obtained an '**F'** grade in any subject shall be deemed to have '**failed'** and is required to reappear as a 'supplementary student' in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.

To a student who has not appeared for an examination in any subject, '**Ab'** grade will be allocated in that subject, and he is deemed to have '**Failed'**. A student will be required to reappear as a 'supplementary student' in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.

A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.

A student earns Grade Point (GP) in each subject/ course, on the basis of the letter grade secured in that subject/ course. The corresponding 'Credit Points' (CP) are computed by multiplying the grade point with credits for that particular subject/ course.

Credit Points (CP) = Grade Point (GP) x Credits …. For a course

A student passes the subject/ course only when **GP $\geq$ 5 ('C' grade or above)**

The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points (ΣCP) secured from all subjects/ courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA is thus computed as

GPA = $\{\sum_{i=1}^{N} C_i G_i\} / \{\sum_{i=1}^{N} C_i\}$ … For each Semester,

where 'i' is the subject indicator index (considering all subjects in a semester), 'N' is the no. of subjects '**registered'** for the semester (as specifically required and listed under the course structure of the parent department), C_i is the no. of credits allotted to the i^{th} subject, and G_i represents the grade points (GP) corresponding to the letter grade awarded for that i^{th} subject.

The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all** semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to **two** decimal places. CGPA is thus computed from the I year II semester onwards at the end of each semester as per the formula

CGPA = $\{\sum_{j=1}^{M} C_j G_j\} / \{\sum_{j=1}^{M} C_j\}$ for all S Semesters registered

(i.e., up to and inclusive of S Semesters, S $\geq$ 2),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1^{st} Semester onwards up to and inclusive of the Semester S (obviously M > N), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the jth Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

Illustration of calculation of SGPA:

Course/Subject	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	4 x 8 = 32
Course 2	4	O	10	4 x 10 = 40
Course 3	4	C	5	4 x 5 = 20
Course 4	3	B	6	3 x 6 = 18

Course 5	3	A+	9	3 x 9 = 27
Course 6	3	C	5	3 x 5 = 15
	21			152

SGPA = 152/21 = 7.24

Illustration of Calculation of CGPA up to 3rd Semester:

Semester	Course/ Subject Title	Credits Allotted	Letter Grade Secured	Corresponding Grade Point (GP)	Credit Points (CP)
I	Course 1	3	A	8	24
I	Course 2	3	O	10	30
I	Course 3	3	B	6	18
I	Course 4	4	A	8	32
I	Course 5	3	A+	9	27
I	Course 6	4	C	5	20
II	Course 7	4	B	6	24
II	Course 8	4	A	8	32
II	Course 9	3	C	5	15
II	Course 10	3	O	10	30
II	Course 11	3	B+	7	21
II	Course 12	4	B	6	24
II	Course 13	4	A	8	32
II	Course 14	3	O	10	30
III	Course 15	2	A	8	16
III	Course 16	1	C	5	5
III	Course 17	4	O	10	40
III	Course 18	3	B+	7	21
III	Course 19	4	B	6	24
III	Course 20	4	A	8	32
III	Course 21	3	B+	7	21
	Total Credits	**69**		**Total Credit Points**	**518**

CGPA = 518/69 = 7.51

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8th semester. The CGPA obtained at the end of 8th semester will become the final CGPA secured for entire B.Tech. programme.

For merit ranking or comparison purposes or any other listing, **only** the '**rounded off'** values of the CGPAs will be used.

SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

Passing Standards

A student shall be declared successful or 'passed' in a semester, if he secures a GP ≥ 5 (**'C'** grade or above) in every subject/course in that semester (i.e. when the student gets an SGPA ≥ 5.0 at the end of that particular semester); and he shall be declared successful or 'passed' in the entire undergraduate programme, only when gets a CGPA
≥ 5.00 (**'C'** grade or above) for the award of the degree as required.

After the completion of each semester, a grade card or grade sheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

Declaration of results

Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.

For final percentage of marks equivalent to the computed final CGPA, the following formula may be used.

% of Marks = (final CGPA – 0.5) x 10

Award of Degree

A student who registers for all the specified subjects/ courses as listed in the course structure and secures the required number of 160 credits (with CGPA ≥ 5.0), within 8 academic years from the date of commencement of the first academic year, shall be declared to have '**qualified'** for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.

A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.

A student with final CGPA (at the end of the undergraduate programme) > 8.00, and fulfilling the following conditions - shall be placed in '**First Class with Distinction**'. However, he

(i) Should have passed all the subjects/courses in '**First Appearance'** within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.

(ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.

A student not fulfilling any of the above conditions with final CGPA > 8 shall be placed in **'First Class'.**

Students with final CGPA (at the end of the undergraduate programme) ≥ 7.0 but <

8.00 shall be placed in **'First Class'.**

Students with final CGPA (at the end of the undergraduate programme) ≥ 6.00 but < 7.00, shall be placed in '**Second Class**'.

All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme) ≥ 5.00 but < 6, shall be placed in '**pass class**'.

A student with final CGPA (at the end of the undergraduate programme) < 5.00 will not be eligible for the award of the degree.

Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of '**Gold Medal**'.

Award of 2-Year B.Tech. Diploma Certificate

1. A student is awarded 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (with in 4 years from the date of admission) upto B. Tech. – II Year – II Semester, if the student want to exit the 4-Year B. Tech. program. The student **once opted and awarded for 2-Year UG Diploma Certificate, the student will not be permitted to join** in B. Tech. III Year – I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree.
2. A student may be permitted to take one year break after completion of II Year – II Semester or B. Tech. – III Year – II Semester (with Principal permission through the HOD of the department well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).

Withholding of results

If the student has not paid the fees to the College at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester.

The award or issue of the degree may also be withheld in such cases.

Transitory Regulations

A. For students detained due to shortage of attendance:

1. A Student who has been detained in I year of R18 Regulations due to lack of attendance, shall be permitted to join I year I Semester of R22 Regulations and he is required to complete the study of B.Tech./B. Pharmacy programme within the stipulated period of eight academic years from the date of first admission in I Year.
2. A student who has been detained in any semester of II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of R22 Regulations and is required to complete the study of B.Tech./B. Pharmacy within the stipulated period of eight academic years from the date of first admission in I Year. The R22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory Regulations.
3. For students detained due to shortage of credits: A student of MLR18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of R22 Regulations only after acquiring the required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both MLR18 & R22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The R22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations.

B. For readmitted students in R22 Regulations:

4. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
5. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including R22 Regulations. **There is NO exemption of credits in any case**.
6. If a student is readmitted to R22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in R22 Regulations will be substituted by another subject to be suggested by the Concerned BOS.

Student Transfers

There shall be no branch transfers after the completion of admission process.

There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.

The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.

The transferred students from other Universities/Institutions to College who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter issued by the University.

The college have to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

Malpractice Prevention Committee

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

a) Controller of examinations - Chairman
b) Addl. Controller of examinations.- Member Convener
c) Subject expert - member
d) Head of the department of which the student belongs to. - Member
e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

Scope

The academic regulations should be read as a whole, for the purpose of any interpretation.

In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of Academic Council is final.

The College may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the college authorities.

Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

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ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY SCHEME) FROM THE A.Y. 2023-24

1. **Eligibility for the award of B.Tech Degree (LES)**

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The student shall register for 120 credits and secure 120 credits with CGPA $\geq$ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.

3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.

4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).

5. **Promotion rule**

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to second year second semester	Regular course of study of second year first semester.
2	Second year second semester to third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to third year second semester	Regular course of study of third year first semester.
4	Third year second semester to fourth year first semester	i) Regular course of study of third year second semester. ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Fourth year first semester to fourth year second semester	Regular course of study of fourth year first semester.

6. All the other regulations as applicable to B. Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.

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Malpractices Rules
Disciplinary Action For / Improper Conduct in Examinations

	Nature of Malpractices/ Improper conduct	Punishment
	If the student:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination).	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other student orally or by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or anyother form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled
3.	Impersonates any other student in connection with the examination.	The student who has impersonated shall be expelled from examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the chief superintendent/COE/ACoE/any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

8.	Possesses any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.
9.	If student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat. Person(s) who do not belong to the college will be handed over to the police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations andproject work of that semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Principal for further action to award a suitable punishment.	

Malpractices identified by squad or special invigilators

1. Punishments to the students as per the above guidelines.
2. Punishment for staff: (if the squad reports that the staff is also involved in encouraging malpractices)
   a. A show-cause notice shall be issued to the staff.
   b. Impose a suitable fine on the staff.

COURSE STRUCTURE

MLR INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
COURSE STRUCTURE – R22

I YEAR I SEMESTER

Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	0	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A6HS01	English for skill Enhancement	HSMC	3	0	0	3	40	60	100
A6BS11	Engineering Chemistry	BSC	3	1	0	4	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	0	0	3	1.5	40	60	100
A6HS02	English Language and Communication skills Lab	HSMC	0	0	3	1.5	40	60	100
A6EC04	Introduction to Internet of Things	ESC	0	0	2	1	40	60	100
A6HS04	Seminar	HSMC	0	0	2	1	50	0	50
A6IT01	Basics of Information Technology	ESC	1	0	0	1	50	0	50
		TOTAL	13	2	10	20	380	420	800

I YEAR II SEMESTER

Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	0	4	40	60	100
A6BS06	Applied Physics	BSC	3	1	0	4	40	60	100
A6EE60	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	40	60	100
A6ME02	Engineering Drawing	ESC	1	0	3	2.5	40	60	100
A6EC03	Electronic Devices and Applications	ESC	2	0	0	2	40	60	100
A6BS07	Applied Physics Lab	BSC	0	0	3	1.5	40	60	100
A6CS04	Python Programming Lab	ESC	0	0	3	1.5	40	60	100
A6ME04	Engineering Work Shop	ESC	0	0	3	1.5	40	60	100
A6BS13	Environmental Science	BSC	2	0	0	0	50	0	50
		TOTAL	14	2	12	20	370	480	850

II YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6BS03	Computer Oriented Statistical Methods	BSC	3	1	0	4	40	60	100
A6CS08	Discrete Mathematics	BSC	3	0	0	3	40	60	100
A6CS05	Data Structures	ESC	3	0	0	3	40	60	100
A6CS09	Database Management Systems	PCC	3	0	0	3	40	60	100
A6HS08	Business Economics and Financial Analysis	HSMC	3	0	0	3	40	60	100
A6CS06	Data Structures Lab	ESC	0	0	3	1.5	40	60	100
A6CS10	Database Management Systems Lab	PCC	0	0	3	1.5	40	60	100
A6IT04	Skill Development Course (HTML,CSS,JAVASCRIPT,BOOTSTRAP)	PCC	0	0	2	1	0	100	100
		TOTAL	15	1	8	20	280	520	800
Mandatory Course									
A6HS05	**Gender sensitization**		0	0	2	0	50	0	50

II YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6CS28	Digital Electronics and Computer Organization	ESC	3	0	0	3	40	60	100
A6CS15	Design and Analysis of Algorithms	PCC	3	0	0	3	40	60	100
A6IT02	Object Oriented Programming through java	PCC	3	0	0	3	40	60	100
A6CS13	Software Engineering	PCC	3	0	0	3	40	60	100
A6CS11	Operating systems	PCC	3	0	0	3	40	60	100
A6IT03	Object Oriented Programming through java Lab	PCC	0	0	3	1.5	40	60	100
A6CS12	Operating systems Lab	PCC	0	0	3	1.5	40	60	100
A6CT01	Real Time Research Project/Societal Related Project	PWC	0	0	4	2	0	100	100
	TOTAL		15	0	10	20	280	520	800
Mandatory Course									
A6HS06	Constitution of India		3	0	0	0	50	0	50

III YEAR I SEMESTER

Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6IT10	Full Stack Development	PCC	3	1	0	4	40	60	100
A6IT11	Automata and Compiler Design	PCC	3	0	0	3	40	60	100
A6IT13	Cloud & DevOps	PCC	3	0	0	3	40	60	100
A6IT12	Data Communication & Computer Networks	PCC	3	0	0	3	40	60	100
PEC	Professional Elective– I	PEC	3	0	0	3	40	60	100
A6IT14	Full Stack Development Lab	PCC	0	0	3	1.5	40	60	100
A6IT16	Cloud & DevOps Lab	PCC	0	0	3	1.5	40	60	100
A6IT17	Skill Development Course (Flutter)	PCC	0	0	0	1	-	100	100
		TOTAL	15	1	6	20	280	520	800
Mandatory Course									
A6HS13	Intellectual Property Rights		3	0	0	0	50	0	50

III YEAR II SEMESTER

Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AI06	Machine Learning	PCC	3	0	0	3	40	60	100
A6DS06	Big Data Technologies	PCC	3	0	0	3	40	60	100
PEC	Professional Elective– II	PEC	3	0	0	3	40	60	100
PEC	Professional Elective –III	PEC	3	0	0	3	40	60	100
OEC	Open Elective-I	OEC	3	0	0	3	40	60	100
A6AI09	Machine Learning Lab	PCC	0	0	3	1.5	40	60	100
A6DS07	Big Data Technologies Lab	PCC	0	0	3	1.5	40	60	100
A6CT02	Industrial Oriented Mini Project	PWC	0	0	4	2	0	100	100
A6BS13	Environmental Science		2	0	0	0	50	0	50
		TOTAL	17	1	10	20	280	520	800

Note : Environmental Science in III-II should be registered by Lateral Entry students only

IV YEAR I SEMESTER									
Code	**Course**	**Category**	**Periods per Week**			**Credits**	**Scheme of Examination Maximum Marks**		
			L	**T**	**P**		**CIE**	**SEE**	**Total**
A6CY06	Cryptography and Network Security	PCC	3	0	0	3	40	60	100
A6IT32	Software Project Management	PCC	3	0	0	3	40	60	100
PEC	Professional Elective -IV	PEC	3	0	0	3	40	60	100
PEC	Professional Elective-V	PEC	3	0	0	3	40	60	100
OEC	Open Elective-II	OEC	3	0	0	3	40	60	100
A6CY08	Cryptography and Network Security Lab	PCC	0	0	2	1	40	60	100
A6CT03	Major Project Phase -1	PWC	0	0	8	4	40	60	100
		TOTAL	**15**	**-**	**10**	**20**	**280**	**420**	**700**

IV YEAR II SEMESTER									
Code	**Course**	**Category**	**Periods per Week**			**Credits**	**Scheme of Examination Maximum Marks**		
			L	**T**	**P**		**CIE**	**SEE**	**Total**
A6HS15	Organizational Behavior	PCC	3	0	0	3	40	60	100
PEC	Professional Elective-VI	PEC	3	0	0	3	40	60	100
OEC	Open Elective-III	OEC	3	0	0	3	40	60	100
A6CT04	Major Project Phase-2 and seminars	PWC	0	0	16	9+2=11	40	60	100
		TOTAL	**9**	**-**	**16**	**20**	**160**	**240**	**400**

I B.TECH I SEMESTER
SYLLABUS

LINEAR ALGEBRA AND CALCULUS

I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS01	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil				Total Classes: 52		

COURSE OBJECTIVES

To learn

1. Concept of Rank of a matrix, Consistency and solving system of linear equations.
2. Concept of eigen values, eigen vectors and diagonalization of the matrix.
3. The concept of differential equations and solve them using appropriate methods.
4. Evaluate multiple integrals and improper integrals
5. The partial derivatives of several variable functions.

COURSE OUTCOMES

at the end of the course, student will be able to:

1. Solve the system of linear equations using rank of the matrices.
2. Find the Eigen values and Eigen vectors of a matrix
3. Identify the different types of differential equations and solve them using appropriate methods.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

UNIT - I	MATRICES AND THEIR APPLICATIONS	CLASSES: 08

Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew –Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m n and n n linear system of equations by Gauss Elimination, Gauss siedel Method

UNIT - II	EIGEN VALUES, EIGEN VECTORS	CLASSES: 08

Eigen values and Eigen vectors and its properties (without proof), Cayley-Hamilton theorem (Statement and
verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices. Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to Canonical
forms by Orthogonal Transformation.

UNIT - III	ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS	CLASSES: 10

Introduction- Exact and reducible to Exact differential equations-Newton's Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, Sin ax, Cosax, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.

UNIT - IV	MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS	CLASSES: 10
Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integral. Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.		
UNIT - V	CALCULUS OF SEVERAL VARIABLES	CLASSES: 08
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule,Jacobians-functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints).		
TEXT BOOKS		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
REFERENCE BOOKS		
1. G.B.Thomas, calculus and analytical geometry, 9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.		

PROGRAMMING FOR PROBLEM SOLVING

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS02	ESC	3	0	0	3	40	60	100
Contact Classes: 64	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 64			

COURSE OBJECTIVES

1. To familiarize with the syntax and semantics of C programming language.
2. To learn the usage of structured programming approach in solving problems.
3. To use arrays, pointers, strings and structures in solving problems.
4. To understand how to solve problems related to matrices, Searching and sorting.
5. To understand how to use files to perform read and write operations.

COURSE OUTCOMES

1. Apply algorithmic thinking to understand, define and solve problems
2. Develop computer programs using programming constructs and control structures and to use arrays to develop C programs
3. Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems.
4. Use Strings and structures to formulate algorithms and programs.
5. Use FILE to perform read and write operations.

UNIT - I	INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING& INTRODUCTION TO C LANGUAGE	CLASSES: 12

Algorithm -Definition, Characteristics of Algorithm.Constituents of algorithms**: -** Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum and Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples.
Introduction To C Language: Structure of C Program, Data Types, Input and Output statements, Operators, Precedence and Associativity of operators , Evaluation of Expressions, Type Conversions In Expressions.

UNIT - II	CONTROL STRUCTURES AND ARRAYS	CLASSES: 15

Control structures: Decision statements; if and switch statement; **Loop control statements:** while, for and do while loops, Jump statements: break, continue, goto statements.
Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays

UNIT - III	FUNCTIONS AND POINTERS	CLASSES: 17

Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions**:** Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc.. Storage classes. **Pointers:** Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.		
UNIT - IV	**STRINGS AND USER DEFINED DATA TYPES**	**CLASSES: 10**
Strings: Array of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings **Structures and Unions**: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.		
UNIT - V	**FILE HANDLING,SEARCHING AND SORTING**	**CLASSES: 10**
File Handling: Command Line Arguments, File Modes, Basic File Operations Read, Write and Append, Example Programs. Random Access Using fseek, ftell and rewind Functions. **Basic Searching And Sorting Algorithms:** Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.		

TEXT BOOKS

1. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)
2. Byron Gottfried, “Programming with C”, Schaum's Outlines Series, McGraw Hill Education, 3rdedition, 2017.
3. Programming in C E. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990

REFERENCE BOOKS

1. W. Kernighan Brian, Dennis M. Ritchie, “The C Programming Language”, PHI Learning, 2nd Edition, 1988.
2. YashavantKanetkar, “Exploring C”, BPB Publishers, 2nd Edition, 2003.
3. Schildt Herbert, “C: The Complete Reference”, Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, “Programming with C”, Universities Press, 2nd Edition, 2012.
4. Dey Pradeep, Manas Ghosh, “Computer Fundamentals and Programming in C”, Oxford University Press, 2nd Edition, 2006.
5. Stephen G. Kochan, “Programming in C”, Addison-Wesley Professional, 4th Edition, 2014.

ENGLISH FOR SKILL ENHANCEMENT

I B. TECH- I SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6HS01	**HSMC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	0	0	3	40	60	100
Contact Classes: 64	**Tutorial Classes: NIL**	**Practical Classes: NIL**				**Total Classes: 64**		

COURSE OBJECTIVES

The course will enable the students to:

1. Develop language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

COURSE OUTCOMES

1. Will be able to acquire language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Use language components to communicate effectively in formal and informal situations

UNIT - I	**'Toasted English' by R. K. Narayan**	**CLASSES: 08**
Vocabulary	: The Concept of Word Formation -The Use of Prefixes and Suffixes – Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives – Synonyms and Antonyms	
Grammar	: Identifying Common Errors in Writing with Reference to Articles and Prepositions	
Reading:	: Reading and Its Importance- Techniques for Effective Reading	
Writing	: Sentence Structures – Use of Phrases and Clauses in Sentences – Importance of Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents.	
UNIT - II	**'Appro JRD' by Sudha Murthy**	**CLASSES: 09**

Vocabulary	: Words Often Misspelt–Homophones, Homonyms and Homographs	
Grammar	: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement	
Reading	: Sub-Skills of Reading –Skimming and Scanning– Exercises for Practice	
Writing	: Nature and Style of Writing-Defining/Describing People, Objects, Places and Events– Classifying-Providing Examples or Evidence.	
UNIT - III	**'Lessons from Online Learning' by F.Haider Alvi, Deborah Hurst et al**	**CLASSES: 09**
Vocabulary	: Words Often Confused– Words from Foreign Languages and their Use in English.	
Grammar	:Identifying Common Errorsin Writing with Reference to Misplaced Modifiers and Tenses	
Reading	:Sub-Skills of Reading – Intensive Reading and Extensive Reading –Exercises for Practice	
Writing	:Format of a Formal Letter - Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV/ Resume	
UNIT - IV	**Art and Literature**	**CLASSES: 08**
Vocabulary	: Standard Abbreviations in English	
Grammar	: Redundancies and Clichés in Oral and Written Communication	
Reading	:Survey, Question, Read, Recite and Review (SQ3RMethod) – Exercises for Practice	
Writing	:Writing Practices- Essay Writing-Writing Introduction and Conclusion-Précis Writing	
UNIT - V	**Go, Kiss the World' by Subroto Bagchi**	**CLASSES: 08**
Vocabulary	: Technical Vocabulary and their Usage	
Grammar	: Common Errors in English (*Covering all the other aspects of grammar which were not covered in the previous units*)	
Reading	: Reading Comprehension-Exercises for Practice	
Writing	: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report	
TEXT BOOKS		
1. "English: Language, Context and Culture" by Orient BlackSwan Pvt. Ltd, Hyderabad.2022.Print.		
REFERENCE BOOKS		

1. Effective Academic Writing by Liss and Davis(OUP)
2. Richards,JackC.(2022)InterchangeSeries.Introduction,1,2,3.CambridgeUniversityPress
3. Wood,F.T.(2007). Remedial English Grammar.Macmillan.
4. Chaudhuri,SantanuSinha.(2018).LearnEnglish:AFunBookofFunctionalLanguage,GrammarandVocabulary.(2nd ed.,).SagePublicationsIndiaPvt.Ltd.
5. (2019).Technical Communication. Wiley India Pvt. Ltd.
6. Vishwamohan,Aysha.(2013).EnglishforTechnicalCommunicationforEngineeringStudents.McGraw-Hill Education India Pvt.Ltd.
7. Swan,Michael.(2016).PracticalEnglishUsage.OxfordUniversityPress.FourthEdition

ENGINEERING CHEMISTRY

I B. TECH- I/II SEMESTER: AERO, CSE, CSC, CSD, CSIT, CSM, EEE, ECE, IT, MECH

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6BS11	BSC	3	1	0	4	40	60	100
Contact Classes:50	Tutorial Classes: 0	Practical Classes: 0				Total Classes: 50		

COURSE OBJECTIVES:

1. To bring adaptability to new developments in Engineering Chemistry and to acquire the skills required to become a perfect engineer.
2. To include the importance of water in industrial usage, fundamental aspects of battery chemistry, significance of corrosion it's control to protect the structures.
3. To imbibe the basic concepts of petroleum and its products.
4. To acquire required knowledge about engineering materials like cement, smart materials and Lubricants.

COURSE OUTCOMES:

At the end of the course student will be able to:

1. Understand the basic properties of water and its usage in domestic and industrial purpose.
2. Acquire the basic knowledge of electrochemical procedures related to corrosion
3. Learn the fundamentals and general properties of polymers and other engineering materials.
4. Predict potential applications of chemistry and practical utility in order to become good engineers and entrepreneurs.

UNIT-I	WATER AND ITS TREATMENT	CLASSES: 10

Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break - point chlorination. Defluoridation- Determination of F- ion by ion- selective electrode method.

Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water - Calgon conditioning - Phosphate conditioning - Colloidal conditioning, External treatment methods - Softening of water by ion- exchange processes. Desalination of water – Reverse osmosis.

UNIT-II	BATTERY CHEMISTRY AND CORROSION	CLASSES: 10

Introduction - Classification of batteries- primary, secondary and reserve batteries with examples. Basic requirements for commercial batteries. Construction, working and applications of: Zn-air and Lithium ion battery, Applications of Li-ion battery to electrical vehicles. Fuel Cells- Differences between battery and a fuel cell, Construction and applications of Methanol Oxygen fuel cell and Solid oxide fuel cell. Solar cells - Introduction and applications of Solar cells.

Corrosion: Causes and effects of corrosion – theories of chemical and electrochemical corrosion – mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection – Sacrificial anode and impressed current methods.

UNIT-III	POLYMERIC MATERIALS	CLASSES: 10
Definition – Classification of polymers with examples – Types of polymerization – addition (free radical addition) and condensation polymerization with examples – Nylon 6:6, Terylene **Plastics:** Definition and characteristics- thermoplastic and thermosetting plastics, Preparation, Properties and engineering applications of PVC and Bakelite, Teflon, Fiber reinforced plastics (FRP). **Rubbers:** Natural rubber and its vulcanization. **Elastomers:** Characteristics –preparation – properties and applications of Buna-S, Butyl and Thiokol rubber. **Conducting polymers:** Characteristics and Classification with examples-mechanism of conduction in trans-polyacetylene and applications of conducting polymers. **Biodegradable polymers:** Concept and advantages - Polylactic acid and poly vinyl alcohol and their applications.		
UNIT-IV	**ENERGY SOURCES**	**CLASSES: 10**
Introduction, Calorific value of fuel – HCV, LCV- Dulongs formula. Classification- solid fuels: coal – analysis of coal – proximate and ultimate analysis and their significance. Liquid fuels – petroleum and its refining, cracking types – moving bed catalytic cracking. Knocking – octane and cetane rating, synthetic petrol - Fischer-Tropsch's process; Gaseous fuels – composition and uses of natural gas, LPG and CNG, Biodiesel – Transesterification, advantages.		
UNIT-V	**ENGINEERING MATERIALS**	**CLASSES: 10**
Cement: Portland cement, its composition, setting and hardening. Smart materials and their engineering applications Shape memory materials- Poly L- Lactic acid. Thermo response materials- Polyacryl amides, Poly vinyl amides **Lubricants:** Classification of lubricants with examples-characteristics of a good lubricants – mechanism of lubrication (thick film, thin film and extreme pressure)- properties of lubricants: viscosity, cloud point, pour point, flash point and fire point.		

TEXT BOOKS:

1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010
2. Engineering Chemistry by Rama Devi, Venkata Ramana Reddy and Rath, Cengage learning, 2016
3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K. Shashikala, Pearson Publications, 2021.
4. 4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications

REFERENCE BOOKS:

1. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi (2015)
2. 2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi (2011)

PROGRAMMING FOR PROBLEM SOLVING LAB

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS03	ESC	0	0	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36				Total Classes: 36		

COURSE OBJECTIVES

1. To be familiarize with flowgorithm to solve simple problems
2. To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
3. To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings pointers and structures.

COURSE OUTCOMES

At the end of the course, student will be able to

1. Solve simple mathematical problems using Flowgorithm.
2. Correct syntax errors as reported by the compilers and logical errors encountered at run time Develop programs by using decision making and looping constructs.
3. Implement real time applications using the concept of array, pointers, functions and structures.
4. Solve real world problems using matrices, searching and sorting.

LIST OF EXPERIMENTS

WEEK - 1 INTRODUCTION TO FLOGORITHM

1. Installation and working of Flowgorithm Software.
2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc.
3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle).
4. Draw a flowchart to find the sum of individual digits of a 3 digit number
5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.
6. Draw a flowchart to find roots of a quadratic equation.
7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd
8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene triangle
9. Draw a flowchart to check whether a given number is palindrome or not.

WEEK - 2 BASIC DATA TYPES

1. Write a C program to find division of 1 integer 1 float numbers.
2. Write a C program to find division of 2 integer numbers.

3. Write a C program to find average of (1int,1float) numbers.
4. Write a C Program to Swap Numbers without Using Temporary Variable
5. Write a C Program to Swap two Numbers Using Temporary Variable
6. Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line.
   a) (x+y+z) / (x-y-z)
   b) (x+y+z) / 3
   c) (x+y) * (x-y) * (y-z)

WEEK - 3	OPERATORS

1. Write a C program to convert temperature from Fahrenheit to Celsius and vice versa (c=(fh-32)/1.8
2. Write a C program to find area and perimeter of a circle. (area=∏r2 perimeter=2∏r)
3. Write a C program to calculate area and perimeter of a right angled triangle.
   a. (Area=1/2*b*h perimeter=w+h+sqrt(w*w+h*h)
4. Find the sum of natural numbers 1 to n.(read n as input) (Use formula sum=n(n+1)/2
5. Write a C program to calculate Simple interest (SI=PTR/100)
6. Write a C program to calculate area and perimeter of a rectangle. Area=l*b Perimeter=2* (l+b)
7. Write a C program to calculate the value of the third angle of a triangle if two angles are given as input.(a+b+c=180)
8. Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. (Amt=num of units*cost)
9. Write a C Program to calculate area and perimeter of a triangle.
   b. Perimeter=(a+b+c)
   c. s=(a+b+c)/2
   d. Area=sqrt(s*(s-a)(s-b)*(s-c))
10. Write a C program to read five Subject marks and find the average.
11. Write a C program to Calculate Compound interest (CI=p(1+r/100)n

WEEK - 4	CONDITIONAL STATEMENTS

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program which takes two integer operands and one operator form the user(+,-,*,/,% use switch)
3. Write a program to compute grade of students using if else adder. The grades are assigned as followed:
   marks<50 F
   50≤marks< 60 C
   60≤marks<70 B
   70≤marks B+
   80≤marks<90 A
   90≤mars≤ 100 A+
4. Write a C program to whether given year leap year or not.
5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.

WEEK - 5	LOOPING STATEMENTS

1. Write a C program to find Sum of individual digits of given integer
2. Write a C program to generate first n terms of Fibonacci series
3. Write a C Program to find the Sum of Series SUM=1-x2/2! +x4/4!-x6/6!+x8/8!-x10/10!
4. Write a C program to print the Fibonacci sequence up to given value of n.
5. Write a C program to print the multiplication table to the given value of n.
6. Write a C program to check whether given number is palindrome or not
7. Write a C program to check whether given number is perfect or not

WEEK - 6 NESTED LOOPING STATEMENTS

1. Write a C program to generate prime numbers between 1 and n
2. Write a C program to generate Pascal's triangle.
3. Write a C program to generate the following pyramid of numbers.

```
        1
    1   3   1
1   3   5   3   1
```

4. Write a C program to generate the pattern

```
*****
 ****
  ***
   **
    *
```

WEEK - 7 ARRAYS

1. Write a C Program to implement following searching methods
   i. Binary Search
   ii. Linear Search
2. Write a C program to find largest and smallest number in a list of integers
3. Write a C program
   iii. To add two matrices
   iv. To multiply two matrices
4. Write a C program to find Transpose of a given matrix

WEEK - 8 FUNCTIONS

1. Write a C program to find the factorial of a given integer using non recursive functions
2. Write a C program to find GCD of given integers using non recursive functions
3. Write a C Program to find the power of a given number using non recursive functions
4. Write a C program to find sum of natural numbers using non recursive function.
5. Write a C program to reverse a given integer number using non recursive functions
6. Write a C Program to find binary equivalent of a given decimal number using recursive functions.
7. Write a C Program to print Fibonacci sequence using recursive functions.
8. Write a C Program to find LCM of 3 given numbers using recursive functions

9. Write a C program to find the factorial of a given integer using recursive functions
10. Write a C program to print fibonocci series till n terms using recursion.

WEEK - 9	**STRINGS**

1. Write a C program using to Insert a sub string into a given main string from a given position
2. Write a C program using to Delete n characters from a given position in a string
3. Write a C program to determine if given string is palindrome or not
4. Write C Programs to demonstrate the following string handling functions.
   a. strcat()
   b. strcmp()
   c. strrev()
   d. strcpy()
   e. strlen()
   f. strstr()
   g. strncpy()
   h. strncat()
   i. strncmp()

WEEK - 10	**POINTERS**

1. Write a C program to read the elements of 1-d array using pointers and print them in reverse order using pointers.
2. Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference.
3. Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array.
4. Write a C Program to print 2-D array using pointers.

WEEK - 11	**STRUCTURES**

1. Write a C Program using functions to
   a) Reading a complex number
   b) Writing a complex number
   c) Add two complex numbers
   d) Multiply two complex numbers

Note: represent complex number using structure

2. Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.

WEEK - 12	**FILES**

1. Write a C program to read and print the content of a file.
2. Write a C program copy the content of one file to another file
3. Write a C program to merge two file into third file.
4. Write a C Program to find the number of lines in a text file

TEXT BOOKS

1. Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014 Mar 27.
2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) YashavantKanetkar, “Let Us C”, BPB Publications, New Delhi, 13th Edition, 2012.

REFERENCE BOOKS

1. Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018
2. King KN, “C Programming: A Modern Approach”, Atlantic Publishers, 2nd Edition, 2015.
3. Kochan Stephen G, “Programming in C: A Complete Introduction to the C Programming Language”, Sam‘s Publishers, 3rd Edition, 2004.
4. Linden Peter V, “Expert C Programming: Deep C Secrets”, Pearson India, 1st Edition, 1994.

ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB

I B. TECH- I SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6HS02	**HSMC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		0	0	3	1.5	40	60	100
Contact Classes: 00	**Tutorial Classes: 00**	**Practical Classes: 39**				**Total Classes:39**		

COURSE OBJECTIVES

The course should enable the students to:

1. To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning
2. To sensitize the students to the nuances of English speech sounds, word accent, intonation and rhythm
3. To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
4. To improve the fluency of students in spoken English and neutralize the impact of dialects.
5. To train students to use language appropriately for public speaking, group discussions and interviews

COURSE OUTCOMES

Students will be able to:

1. Acquire the skill of independent language learning
2. Overcome with the nuances of English speech sounds, word accent, intonation and rhythm.
3. Obtain the consistent accent and intelligibility in pronunciation.
4. Neutralize the impact of dialects.
5. Apply language appropriately public speaking, group discussions and interviews

English Language and Communication Skills Lab (ELCS) shall have two parts:

a. **Computer Assisted Language Learning (CALL) Lab**
b. **Interactive Communication Skills (ICS) Lab**

Listening Skills

Objectives

1. To enable students develop their listening skills to appreciate its role in the LSRW skills approach to language and improve their pronunciation.
2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions.

Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.

- Listening for general content
- Listening to fill up information
- Intensive listening
- Listening for specific information

Speaking Skills

Objectives

1. To involve students in speaking activities in various contexts
2. To enable students express themselves fluently and appropriately in social and professional contexts

- Oral practice

EXERCISE-I

CALL Lab:
Understand: Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening.
Practice: Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs- Consonant Clusters- Past Tense Marker and Plural Marker- *Testing Exercises*
ICS Lab:
Understand: Spoken vs. Written language- Formal and Informal English.
Practice: Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others.

EXERCISE-II

CALL Lab:
Understand: Structure of Syllables – Word Stress– Weak Forms and Strong Forms – Stress pattern in sentences – Intonation.
Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms- Stress pattern in sentences – Intonation - *Testing Exercises*
ICS Lab:
Understand: Features of Good Conversation – Strategies for Effective Communication.
Practice: Situational Dialogues – Role Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.

EXERCISE-III	
CALL Lab: Understand: Errors in Pronunciation-Neutralising Mother Tongue Interference (MTI). Practice: Common Indian Variants in Pronunciation – Differences between British and American Pronunciation -Testing Exercises **ICS Lab:** Understand: Descriptions- Narrations- Giving Directions and Guidelines – Blog Writing Practice: Giving Instructions – Seeking Clarifications – Asking for and Giving Directions – Thanking and Responding – Agreeing and Disagreeing – Seeking and Giving Advice – Making Suggestions.	
EXERCISE-IV	
CALL Lab: Understand: Listening for General Details. Practice: Listening Comprehension Tests - Testing Exercises **ICS Lab:** Understand: Public Speaking – Exposure to Structured Talks - Non-verbal Communication- Presentation Skills.	
EXERCISE-V	
CALL Lab: *Understand:* Listening for Specific Details. *Practice:* Listening Comprehension Tests -*Testing Exercises* **ICS Lab**: *Understand:* Group Discussion *Practice:* Group Discussion	
REFERENCE BOOKS	

1. (2022). *English Language Communication Skills – Lab Manual cum Workbook.* Cengage Learning India Pvt. Ltd.
2. Shobha, KN &Rayen, J. Lourdes. (2019). *Communicative English – A workbook.* Cambridge University Press
3. Kumar, Sanjay & Lata, Pushp. (2019). *Communication Skills: A Workbook.* Oxford University Press
4. Board of Editors. (2016). *ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities.*Orient Black Swan Pvt. Ltd.
5. Mishra, Veerendra et al. (2020). *English Language Skills: A Practical Approach.* CambridgeUniversity Press

INTRODUCTION TO INTERNET OF THINGS

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC04	ESC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	40	60	100
Contact Classes: 00	Tutorial Classes: 00	Practical Classes: 24				Total Classes:24		

COURSE OBJECTIVES

1. To develop basic programming skills through graphical programming
2. To learn hardware interfacing and debugging techniques
3. To design and develop android apps

COURSE OUTCOMES

At the end of the course, student will be able to the algorithms for simple problems

1. CO 1: Able to demonstrate various sensor interfacing using Visual Programming Language.
2. CO 2: Able to analyze various Physical Components.
3. CO 3: Able to demonstrate Wireless Control of Remote Devices.
4. CO 4: Able to design and develop Mobile Application which can interact with Sensors

INTRODUCTION TO IOT

1. Introduction to basic electronic components and digital electronic
2. Introduction to sensors and Actuators
3. Introduction to microcontroller
4. Introduction to Arduino IDE

LIST OF EXPERIMENTS

WEEK - 1	Blinking of LED with different delays
WEEK - 2	Digital I/O Interface [IR Sensor, PIR Sensor]
WEEK - 3	Analog Interface [ADC, Temperature Sensor]
WEEK - 4	Motor speed And Direction control
WEEK - 5	Serial Communication
WEEK - 6	Wireless Interface –Bluetooth & Wi-Fi Technologies
WEEK - 7	Wireless Control of wheeled robot
WEEK - 8	Smart Home Android App Development

REFERENCES

1. Sylvia Libow Martinez, Gary S Stager, Invent To Learn: Making, Tinkering, and Engineering in the Classroom, Constructing Modern Knowledge Press, 2016
2. Michael Margolis, Arduino Cookbook, Oreilly, 2011

SEMINAR

I B. TECH- I SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
		L	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
A6HS04	**HSMC**	0	0	2	1	50	0	50

COURSE OBJECTIVES

Seminar is an important component of learning in an Engineering College, where the student gets acquainted with preparing a report & presentation on a topic.

PERIODICITY / FREQUENCY OF EVALUATION : Twice

PARAMETERS OF EVALUATION:

1. The seminar shall have topic allotted and approved by the faculty.
2. The seminar is evaluated for 50 marks for internal only.
3. The students shall be required to submit the rough drafts of the seminar outputs within one week of the commencement of the class work.
4. Faculty shall make suggestions for modification in the rough draft. The final draft shall be presented by the student within a week thereafter.
5. Presentation schedules will be prepared by Department in line with the academic calendar.

THE SEMINARS SHALL BE EVALUATED IN TWO STAGES AS FOLLOWS:

A. ROUGH DRAFT

In this stage, the student should collect information from various sources on the topic and collate them in a systematic manner. He/ She may take the help of the concerned faculty.

The report should be typed in “MS-Word” file with “calibri” font, with font size of 16 for main heading, 14 for sub-headings and 11 for the body text. The contents should also be arranged in Power Point Presentation with relevant diagrams, pictures and illustrations. It should normally contain 10 to 15 slides, consisting of the followings:

1.	Topic, name of the student & faculty	1 Slide
2.	List of contents	1 Slide
3.	Introduction	1Slides
4.	Descriptions of the topic (point-wise)	6 - 10 Slides
5.	Conclusion	1 - 2 Slides
6.	References/Bibliography	1 Slide

The soft copy of the rough draft of the seminar presentation in MS Power Point format along with the draft report should be submitted to the concerned faculty, with a copy to the concerned HOD within stipulated time.

Diodes: Diode – Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.

The evaluation of the rough draft shall generally be based upon the following.

1	Punctuality in submission of rough draft	2
2	Dress Code	3
3	Resources from which the seminar have been based	2
4	Report , and content of Presentation	5
5	Depth of the students knowledge in the subject	5
6	Reception from Questions	5
7	Time Management, Classroom Dynamic	3
	Total Marks	**25**

After evaluation of the first draft the supervisor shall suggest further reading, additional work and fine tuning, to improve the quality of the seminar work. Within 7 days of the submission of the rough draft, the students are to submit the final draft incorporating the suggestions made by the faculty.

BASICS OF INFORMATION TECHNOLOGY

<table>
<tr><td colspan="11">I B.Tech - I Semester: IT /CSIT</td></tr>
<tr><td>Course Code</td><td>Category</td><td colspan="3">Hours / Week</td><td>Credits</td><td colspan="3">Maximum Marks</td></tr>
<tr><td rowspan="2">A6IT01</td><td rowspan="2">ESC</td><td>L</td><td>T</td><td>P</td><td>C</td><td>CIE</td><td>SEE</td><td>Total</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>50</td><td>0</td><td>50</td></tr>
<tr><td>Contact Classes: 30</td><td>Tutorial Classes: 00</td><td colspan="4">Practical Classes: 00</td><td colspan="3">Total Classes:30</td></tr>
</table>

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Know the working principles of functional units of a basic Computer
2. Understand program development, the use of data structures and algorithms in problem solving.
3. Know the need and types of operating system, database systems.
4. Understand the significance of networks, internet, WWW and cyber security.
5. Understand Autonomous systems, the application of artificial intelligence.

UNIT-I	Computer Applications & Basics	Classes: 6

Introduction, Basic Applications of Computer, Components of Computer, Computer Hardware & Software, Input Devices, Output Devices, Central Processing Unit, Modern's Processor, Computer Chip, Computer Ports

Computer Memory: Introduction to computer Memory, Types of Memory, Register memory, Cache memory, Primary memory, Secondary memory, Memory Units

UNIT-II	Computer Operating System	Classes: 6

Basics of Operating System, Linux, Windows, MacOS, Android, Task Icons, Bars, System Settings, Setting Date & Time, File Management, Process, Multitasking, Graphical User Interfaces, and Command Line Interface, MS-DOS Commands.

Windows based applications: MS Office.

Graphics and Multimedia: Getting Images into Your Computer, Graphics Software Paint Programs, Draw Programs, Photo-Editinq Program's, Computer-Aided Design Proqrams,3-D and Animation Software.

UNIT-III	Introduction to Computer Networks	Classes: 6

Internet, Internet's History, www, URL, DNS ,Layers of OSI Model, Routers, Types of Networks, TCP/IP, UDP/IP IP address, Dial-up Connections, High-Speed Broadband Connections, Integrated Services Digital Network (1DSN) Service, Cable Modem Service, Firewalls, Networks Interface cards, ,Servers in Computer Network, Applications of Servers.

UNIT-IV	Programming and Databases	Classes: 6

Program: Introduction, Purpose of Program, Level of programming languages, Types of Programming languages, HTML, Fundamentals of Scripting languages

Data Structures: Introduction to Data Structures, categories of Data structures, Basic Operations of Data structures.

UNIT-V	Fundamentals of Design and Technologies	Classes: 6

WordPress: Creating a WordPress Site (Installing WordPress)

Information Technology: Elements of Information Technology, Telecommunication, Fiber Optics, Integrated Service Digital Network (ISDN), Blue-Gene Computer, Cloud Computing, Cyber Crime & Cyber Security, Artificial Intelligence, Machine Learning, IoT, Block Chain, Electronic Commerce.

Text Books:

1. Introduction to computers, Peter Norton, 8th Edition, Tata McGraw Hill, 2017.

Reference Books:

1. Invitation to Computer Science, G. Michael Schneider, Macalester College, Judith L. Gersting University of Hawaii, Hilo, Contributing author: Keith Miller University of Illinois, Springfield. Fundamentals of Computers, Reema Thareja, Oxford Higher Education, Oxford University Press.
2. Computer Fundamentals, Anita Goel, Pearson Education India, 2010.
3. Elements of computer science, Cengage.

I B.TECH II SEMESTER
SYLLABUS

NUMERICAL METHODS AND INTEGRAL TRANSFORMS

I B. TECH- II SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6BS02	**BSC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	1	0	4	40	60	100
Contact Classes: 44	**Tutorial Classes: 08**	**Practical Classes: Nil**				**Total Classes: 52**		

COURSE OBJECTIVES

To learn

1. Curve fitting and Interpolation techniques.
2. Numerical techniques.
3. Fourier series for periodic function
4. Laplace transforms
5. Concept and application of Fourier Transforms and Vector differentiation.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Apply Curve fitting and Interpolation techniques.
2. Apply various numerical techniques
3. Find the Fourier series of the periodic functions.
4. Obtain the Laplace tranforms of functions
5. Find Fourier transforms and apply vector differentiation techniques.

UNIT - I	INTERPOLATION AND CURVE FITTING	CLASSES: 08

INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange's interpolation.

CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of theform $y= a e^{b x}$, $y= a x^{b,}$ $y = a b^{x}$ by the method of least squares.

UNIT - II	NUMERICAL TECHNIQUES	CLASSES: 08

ROOT FINDING TECHNIQUES:

Bisection method Regulafalsimethod,Iteration method and Newton Raphson method.

NUMERICAL INTEGRATION :

Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule.

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS:

Taylor's series method – Euler's - modified Euler's Method – Runge-Kutta method.

UNIT - III	FOURIER SERIES	CLASSES:10

Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.

UNIT - IV	LAPLACE TRANSFORMS	CLASSES: 10

Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n- Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.		
UNIT - V	**FOURIER TRANSFORMS AND VECTOR DIFERENTIATION**	**CLASSES: 10**
Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms. Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.		
TEXT BOOKS		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9^{th} Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
REFERENCE BOOKS		
1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.		

APPLIED PHYSICS

I B. TECH- II SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6BS06	**BSC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	1	0	4	40	60	100
Contact Classes: 50	**Tutorial Classes: 08**	**Practical Classes: Nil**				**Total Classes: 58**		

COURSE OBJECTIVES

The course should enable the students to:

1. Learn the basic principles of quantum physics and its applications
2. Understand the formation of energy bands and atomic structure in solids for material classification
3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices
4. Learn the basic principles of laser and optical fiber in information technology
5. Identify the importance of nanoscale and various fabrication and characterization techniques and quantum computations in engineering applications

COURSE OUTCOMES

At the end of the course students will be able to:

1. **Analyze** the microscopic properties of materials using principles of quantum physics for engineering applications
2. **Explain** the behavior of different electronic materials based on the concepts of band theory
3. **Apply** the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation
4. **Gain** the knowledge of production of laser and usage of fibers in fiber optic communication technology
5. **Comprehend** the knowledge of quantum physics in quantum computation for secure information technology

UNIT - I	**QUANTUM PHYSICS**	**CLASSES: 10**

Black body radiation, Stefen-Boltzmann"s law, Planck's radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.

UNIT - II	**ELECTRONIC PROPERTIES OF MATERIALS & BAND THEORY OF SOLIDS**	**CLASSES: 10**

Electronic properties of Materials: Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron.

Band theory of solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac

distribution law, Classification of materials as conductors, insulators and semiconductors.		
UNIT - III	**SEMI-CONDUCTORS & SEMICONDUCTOR DEVICES**	**CLASSES: 08**
Semiconductors: Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications. **Semiconductor Devices**: Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.		
UNIT - IV	**LASER & FIBER OPTICS**	**CLASSES: 12**
Laser: Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO2 laser, Applications of lasers in different fields. **Fiber Optics:** Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.		
UNIT - V	**PHYSICS OF QUANTUM COMPUTING & QUANTUM GATES**	**CLASSES: 10**
Physics of Quantum computing: Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits. **Quantum gates**-Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates-CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.		
TEXT BOOKS		
1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed. 2. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press. 3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy" A Text book of Engineering Physics"- S. Chand Publications, 11th Edition 2019.		
REFERENCE BOOKS		
1. Quantum Physics, H.C. Verma, TBS Publiication, 2nd Edition 2012 2. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11th Edition, 2018. 3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication,2019 4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2nd Edition, 2022		

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

I B. TECH- I SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6EE60	**ESC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	0	0	3	40	60	100
Contact Classes: 64	**Tutorial Classes: NIL**	**Practical Classes: NIL**				**Total Classes: 64**		

COURSE OBJECTIVES

1. Develop fundamentals, including Ohm's law, Kirchhoff's laws and be able to solve for currents, voltages and power in electrical circuits.
2. Develop EMF equation and analyze the operation of DC Machines.
3. Analyze the working principle of Transformer.
4. Discuss the operation of AC Machines.
5. Analyze the operation of PN junction diode and rectifiers.
6. Discuss the operation and characteristics of Transistors.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to :

1. Evaluate current and voltage values in resistive circuits with independent sources.
2. Explain the working of DC machines and solve the numerical problems..
3. Explain the working of AC electrical machines and solve the numerical problems.
4. Analyze the V-I characteristics of PN – junction diode and describe the operation of rectifiers.
5. Analyze the different configurations of Transistors and obtain its characteristics.

UNIT - I	**ELECTRICAL CIRCUITS**	**CLASSES : 12**

Basic definitions-Ohm's Law, types of elements, types of sources , Kirchhoff's Laws – simple problems., series & parallel resistive networks with DC excitation, star to delta and delta to star transformations.

UNIT - II	**DC MACHINES**	**CLASSES : 12**

Principle of Operation of DC Motor, types of DC motor, Torque equation & Losses and problems. DC Generator construction and working Principle, EMF Equation types of generators and problems.

UNIT - III	**AC MACHINES**	**CLASSES : 12**

Working principle and Construction of transformer, Emf Equation & problems. Principle operation of 3-phase induction motor, slip and torque Equation, Torque –slip characteristics & problems.

UNIT - IV	**DIODE AND ITS CHARACTERISTICS**	**CLASSES : 12**

PN JUNCTION DIODE: Operation of PN junction Diode: forward bias and reverse bias, Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers –capacitor filters, inductor filters

UNIT - V	**TRANSISTORS**	**CLASSES : 10**
Bipolar Junction Transistor - NPN & PNP Transistor, CB, CE, CC Configurations and Characteristics.		
TEXT BOOKS		
1. Basic Electrical Engineering by *M.S.Naidu and S.Kamakshaiah* TMH 2. Electronic Devices and circuits by *J.Millman, C.C.Halkias and Satyabrata Jit* 2ed.,		
REFERENCE BOOKS		
1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, "Basic Electrical, Electronics and Computer Engineering",Tata McGraw Hill, Second Edition, (2006). 2. Nagsarkar T K and Sukhija M S, "Basics of Electrical Engineering", Oxford press (2005). 3. Mehta V K, "Principles of Electronics", S.Chand & Company Ltd, (1994). 4. Mahmood Nahvi and Joseph A. Edminister, "Electric Circuits", Schaum' Outline Series, McGraw Hill, (2002).		

ENGINEERING DRAWING

I B. TECH- II SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6ME02	**ESC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		1	0	3	2.5	40	60	100
Contact Classes: 64	**Tutorial Classes: NIL**	**Practical Classes: NIL**				**Total Classes: 64**		

COURSE OUTCOMES

At the end of the course the student should be able to:

1. Understand various commands and create drawing in AutoCAD.
2. Construct various engineering curves and know their importance.
3. Prepare orthographic projections of objects by visualizing them in different positions.
4. Solve the problem of projections of planes and solids in different positions.
5. Construct the isometric view into orthographic views and vice versa.

UNIT - I INTRODUCTION TO ENGINEERING DRAWING

Introduction to Engineering Drawing: Principles and their significance.
Introduction to Computer Aided Drafting: Initial Setup Commands, Draw Commands, modify commands, 2D Drawings-Simple Exercises.

UNIT - II ENGINEERING CURVES

Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only), Involutes.

UNIT - III ORTHOGRAPHIC PROJECTIONS OF POINTS AND LINES

Principles of Orthographic Projections –Projections of points.
Projection of lines inclined to both planes (First angle projection only).

UNIT - IV PROJECTIONS OF PLANES AND SOLIDS

Projections of Planes: Projections of regular planes inclined to one plane.
Projection of Solids**:** Solids inclined to one plane (Regular solids).

UNIT - V ISOMETRIC VIEW AND ORTHOGRAPHIC VIEWS - CONVERSION

Isometric view: Drawing Isometric circles, Dimensioning Isometric Objects.
Conversion of Isometric view to Orthographic views and Orthographic to isometric view.

TEXT BOOKS

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication.

REFERENCE BOOKS

1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka "Engineering Graphics with AutoCAD" PHI publications.
2. Narayana, K.L. & P Kannaiah, Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. & Rana B.C., Engineering Drawing and Computer Graphics, Pearson Education.

ELECTRONIC DEVICES AND APPLICATIONS

I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC03	ESC	L	T	P	C	CIE	SEE	Total
		2	0	0	2	40	60	100

COURSE OBJECTIVES

1. To introduce components such as diodes, BJTs and FETs.
2. To know the applications of devices.
3. To know the switching characteristics of devices.

COURSEOUTCOMES

Upon completion of the Course, the students will be able to:

1. Acquire the knowledge of various electronic devices and the its use on real life.
2. Know the applications of various devices.
3. Acquire the knowledge about the role of special purpose devices and their applications.

UNIT - I DIODES

Diodes: Diode – Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.

UNIT - II DIODE APPLICATIONS

Diode Applications: Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Types of Clampers.

UNIT - III BIPOLAR JUNCTION TRANSISTOR (BJT)

Bipolar Junction Transistor (BJT): Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, Need for Biasing, BJT as Amplifier.

UNIT - IV JUNCTION FIELD EFFECT TRANSISTOR (FET)

Junction Field Effect Transistor (FET): Construction, Principle of Operation, Pinch-Off Voltage, Volt-Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, FET as amplifier.

UNIT - V SPECIAL PURPOSE DEVICES

Special Purpose Devices: Zener Diode - Characteristics, Zener diode as Voltage Regulator, Operation of - SCR, UJT, Photo diode, Solar cell, LED.

TEXT BOOKS	
1.	Jacob Millman - Electronic Devices and Circuits, McGraw Hill Education
2.	Robert L. Boylestead, LouisNashelsky Electronic Devices and Circuits theory, 11^{th} Edition, 2009, Pearson**.**
REFERENCE BOOKS	
1.	Horowitz-Electronic Devices and Circuits, David A. Bell–5^{th}Edition, Oxford.
2.	Chinmoy Saha, Arindam Halder, Debaati Ganguly – Basic Electronics – Principles and Applications, Cambridge, 2018.

APPLIED PHYSICS LAB

I B. TECH- II SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6BS07	**BSC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		0	0	3	1.5	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications
2. To know the Laser and Fibre optic technologies and its applications in real time scenario
3. Understand the electromagnetic properties using experimental knowledge
4. Understand the method of least squares fitting

COURSE OUTCOMES

By the end of the course students will be able to:

1. **Analyze** the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photo diodes, solar cell, and temperature effect on resistance using thermistor.
2. **Identify** the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank's constant using a light source and interference by using Newton's rings
3. **Analyze** the electromagnetic properties of a current carrying coil by using Stewart Gee's experiment
4. **Analyze** the least squares fitting method for data analysis using experimental data of Tensional pendulum

APPLIED PHYSICS LABORATORY
LIST OF EXPERIMENTS

Experiment - 1	**ENERGY GAP OF P-N JUNCION DIODE:** To determine the energy gap of a given semiconductor diode
Experiment – 2	**SOLAR CELL:** To study the V-I and V-P characteristics and determine the fill factor of solar cell
Experiment – 3	**LIGHT EMITTING DIODE:** To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs
Experiment – 4	**HALL EFFECT:** To determine Hall co-efficient and charge carrier concentration of a given semiconductor
Experiment – 5	**PIN PHOTO DIODE:** To study the V-I Characteristics of Photo Diode with respect to intensity of light
Experiment – 6	**OPTICAL FIBRE:** To determine the numerical aperture and acceptance angle of an optical fiber
Experiment – 7	**LASER:** To determine the wavelength of a given laser source by using diffraction grating method

Experiment – 8	**NEWTON'S RINGS:** To determine the radius of curvature of a given Plano convex lens by forming Newton's rings
Experiment – 9	**THERMISTOR:** To study the variation of resistance with respect to temperature using thermistor
Experiment - 10	Understanding the method of least squares - Torsional pendulum as an example
Experiment – 11	**PLANCK'S CONSTANT:** To determine value of planck's constant by measuring radiation in fixed spectral range
Experiment - 12	**STEWART GEE'S EXPERIMENT:** To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux
Note: Students have to perform any 8 experiments	
REFERENCE BOOKS	

1. Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel
2. S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.

PYTHON PROGRAMMING LAB

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS04	ESC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

COURSE OBJECTIVES

1. To understand the problem-solving approaches.
2. To learn the basic programming constructs in Python.
3. To practice various computing strategies for Python-based solutions to real world problems.
4. To use Python data structures – lists, tuples, dictionaries.
5. To do input/output with files in Python

COURSE OUTCOMES

1. Write, test, and debug simple Python programs.
2. Implement Python programs with conditions and loops.
3. Develop Python programs step-wise by defining functions and calling them.
4. Use Python lists, tuples, dictionaries for representing compound data.
5. Read and write data from/to files in Python

LIST OF PROGRAMS

WEEK - 1

a) Write a program to perform different Arithmetic Operations on numbers in Python
b) Write a Python program which accepts the radius of a circle from the user and compute the area.
c) Write a Python program to get the Python version you are using.
d) Write a Python program that accepts an integer (n) and computes the value of n+nn+nnn

WEEK - 2

a) Write a Python program to convert temperatures to and from Celsius, Fahrenheit. [Formula: c/5 = f-32/9] ‘
b) Write a python script to print the current date in the following format “Sun May 29 02:26:23 IST 2017”
c) A library charges a fine for every book returned late. For first 6 days the fine is 50 paisa, for 10-15 days fine is one rupee and above 15 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled.
d) Write a python program to accept the number of days the member is late to return the book and display the fine or the appropriate message.

WEEK - 3	
a) Write a python function to find largest of three numbers. b) Write a Python function that prints prime numbers in between 50 and 100. c) Write a python program to find factorial of a number using Recursion. d) Write a function that receives marks received by a student in 6 subjects and returns the average and percentage of these marks. Call this function from main() and print the result in main	
WEEK - 4	
Write a program to demonstrate working with Strings and string operations.	
WEEK - 5	
Write a program to demonstrate working with dictionaries in python	
WEEK - 6	
Write a program to demonstrate working with tuples and List in python.	
WEEK - 7	
a) Write a script named hellow.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file. b) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order	
WEEK - 8	
Write python programs to demonstrate class & object, static and instance method implementation.	
WEEK - 9	
Write python programs to demonstrate Inheritance and Polymorphism	
WEEK - 10	
Write python programs to demonstrate Exception Handling in python	
WEEK - 11	
Write python programs to demonstrate NumPy library and supporting functions.	
WEEK - 12	
a) Draw an Olympic Symbol in Python using Turtle b) Develop a simple login page using GUI Tkniter	

TEXT BOOKS	
1.	Allen B. Downey, Think Python : How to Think like a Computer Scientist, 2nd Edition, Oâ€™Reilly Publishers, 2016.
2.	Karl Beecher, Computational Thinking: A Beginners Guide to Problem Solving and Programming, 1st Edition, BCS Learning and Development Limited,2017.
REFERENCE REFERENCES	
1.	Paul Deitel and Harvey Deitel, Python for Programmers, Pearson Education, 1st Edition, 2021.
2.	G Venkatesh and Madhavan Mukund, Computational Thinking: A Primer for Programmers and Data Scientists, 1st Edition, Notion Press, 2021.
3.	John V Guttag, Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Dataâ€˜â€˜, Third Edition, MIT Press, 2021
4.	Eric Matthes, Python Crash Course, A Hands – on Project Based Introduction to Programming, 2nd Edition, No Starch Press, 2019. https://www.python.org/

ENGINEERING WORK SHOP

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME04	ESC	0	0	3	1.5	40	60	100

COURSE OUTCOMES

At the end of the course the student should be able to:

1. Assemble the different components
2. Identify and apply suitable tools for different trades of Engineering processes.
3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. Apply basic electrical engineering knowledge for house wiring practice.
5. Learn the safety precautions for various operations in basic trades.

WEEKS	TRADES FOR EXERCISES
	Fitting
WEEK 1	Filing Four Sides of MS Work piece
WEEK 2	L Fit
	Carpentry
WEEK 3	Half Lap Joint
WEEK 4	Dove Tail Joint
	Tin Smithy
WEEK 5	Prepare a Rectangular Tray
WEEK 6	Prepare A Square Tin
	Electrical
WEEK 7	Series and Parallel Connection One Way Switch
WEEK 8	Two Way Switch Connection Stair Case Wiring
	Electronics
WEEK 9	Soldering - Series Connection & Parallel Connection
WEEK 10	Desoldering & Construction of Wheat stone bridge
	TRADES FOR DEMONSTRATION AND EXPOSURE

WEEK 11	Introduction to Black smithy
WEEK 12	Introduction to Plumbing

TEXT BOOKS

1. Work shop Manual - P. Kannaiah/ K.L. Narayana/ Scitech.
2. Workshop Manual / K. Venugopal / Anuradha.

REFERENCE BOOKS

1. Hajra Choudhury S.K., Hajra Choudhury A.K.and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. Workshop Manual / Venkat Reddy/ BSP

ENVIRONMENTAL SCIENCE

I B. TECH- II SEMESTER									
Course Code:	**Category**	**Hours/Week**			**Credits**	**Maximum Marks**		
A6BS13	**BSC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		2	0	0	0	50	0	50
Contact Classes: 30	**Tutorial Classes: 0**	**Practical Classes: 0**				**Total Classes:30**		

COURSE OUTCOMES

On Successful completion of this course, Students will be able to

1. Identify the consequences of human actions on the web of life, global economy and quality of human life.
2. Evaluate the strategies for scientific, social, economic and legal environmental protection.
3. Study the impact of conservation of biodiversity.
4. Analyze the reasons for environmental pollution.
5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.

UNIT - I	ECOSYSTEMS	CLASSES:7

Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids.
Flowof energy, Biogeochemical cycles, Bio accumulation, Bio magnification, eco system value, services and carrying capacity.

UNIT - II	NATURAL RESOURCES	CLASSES: 8

Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources.
Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.

UNIT - III	BIODIVERSITY AND BIOTIC RESOURCES	CLASSES: 5

Bio diversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values.
India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of bio diversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.

UNIT - IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES	CLASSES:5
Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation. **Global Environmental Problems and Global Efforts:** Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions /Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-GoI Initiatives, COP 24, COP25.		
UNIT - V	**ENVIRONMENTAL POLICY, LEGISLATION & EIA**	**CLASSES:5**
Environmental Policy, Legislation & Environmental Impact Assessment (EIA): Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wildlife Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules. **Environmental Impact Assessment:** EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.		

TEXT BOOKS

1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
2. Environmental Studies by R.Rajagopalan, Oxford University Press.

II B.TECH I SEMESTER
SYLLABUS

COMPUTER ORIENTED STATISTICAL METHODS

II B. TECH- I SEMESTER : CSE/IT/CSD/CSC/CSI

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIA	SEE	Total
A6BS03	BSC	3	1	-	4	40	60	100
Contact Classes: 42	Tutorial Classes: 08	Practical Classes:- Nil				Total Classes: 50		

Course Objectives

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. The concept of correlation and regression, covariance and sampling distribution
3. Evaluation of the given data for appropriate test of hypothesis for large samples.
4. Evaluation of the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. To learn the concept of Markov chain, transition probabilities in discrete & continuous time and Stochastic simulation techniques.

UNIT-I	RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS	Classes :8

Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation,.
Binomial, Poisson, Normal distributions -their Properties. Moments about origin, central moments, skewness, Kurtosis and find the mean and variance.

UNIT-II	CORRELATION & REGRESSION AND SAMPLING DISTRIBUTIONS	Classes:9

Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression-Regression Coefficient, The lines of regression.
Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation

UNIT-III	TESTING OF HYPOTHESIS - I	Classes:9

Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I& Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test.
Large sample tests:
(i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances)
(ii) Tests of significance of difference between sample S.D and population S.D.
(iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.

UNIT-IV	TESTING OF HYPOTHESIS-II	Classes: 8

Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit. One

way ANOVA		
UNIT-V	**STOCHASTIC PROCESSES AND MARKOV CHAINS**	**Classes:8**
Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, n-step transition probabilities, Markov chain, Steady state condition, Markov analysis.		
Text Books:		
1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010. 2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall. 3.Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning		
Reference Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. Fundamentals of Mathematical Statistics by S.C. Guptha&V.K. Kapoor, S. Chand 3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press		
Course Outcomes **At the end of the course, student will be able to:** 1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments. 2. Apply the concept of correlation and regression to find covariance and Sampling distribution 3. Evaluate the given data for appropriate test of hypothesis for Large samples. 4. Evaluate the given data for appropriate test of hypothesis for small samples and one way ANOVA 5. Recognize if a given stochastic system with finite number of states is a Markov chain or not and also identify classes of states in Markov chains and characterize the states.		

DISCRETE MATHEMATICS

II B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS08	BSC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To help students understand discrete and continuous mathematical structures
2. To impart basics of relations and functions
3. To facilitate students in applying principles of Recurrence Relations to calculate generating
4. Functions and solve the Recurrence relations
5. To acquire knowledge in graph theory

COURSE OUTCOMES

At the end of the course, student will be able to

1. Analyze and examine the validity of argument by using propositional and predicate calculus
2. Apply basic counting techniques to solve the combinatorial problems
3. Apply sets relations and digraphs to solve applied problems
4. Solve the given recurrence relation using different methods such as substitution, Generating function and characteristics roots equation.
5. Use the basic concepts of graph theory and some related theoretical problems

UNIT - I	MATHEMATICAL LOGIC	CLASSES: 11

Statements and notations, Connectives, Well formed formulas, Truth Tables, Tautology, Equivalence implication, Normal forms, Logical Inference, Rules of inference, Direct Method, Direct Method using CP(Conditional Proof), Consistency, Proof of contradiction, Automatic Theorem Proving.

UNIT - II	RELATIONS	CLASSES: 16

Introduction to set theory**,** Relations, Properties of Binary Relations, Equivalence Relation, Transitive closure, Compatibility and Partial ordering relations, Lattices, Hasse diagram. Functions: inverse Function , Composition of functions. Algebraic Systems, Semi groups and Monoids

UNIT - III	ELEMENTARY COMBINATORICS	CLASSES: 12

Basis of counting, Combinations & Permutations, Enumeration of Combinations and Permutations, Enumeration of Combinations and Permutations With repetitions, Enumerating Permutations with Constrained repetitions, Binomial Coefficients, Binomial and Multinomial theorems, The principles of Inclusion – Exclusion, Pigeon- hole principles and its applications.

UNIT - IV	RECURRENCE RELATION	CLASSES: 11

Generating Functions, Function of Sequences, Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions , The method of Characteristics roots,

UNIT - V	GRAPHS	CLASSES: 10

Basic Concepts, Isomorphism and Subgraphs, Trees and their properties, Spanning Trees- DFS,BFS, Minimal Spanning Trees- Prims, Kruskal’s Algorithm, Planar Graphs, Euler's Formula, Multi graph and Euler circuits, Hamiltonian Graphs, Chromatic number.
TEXT BOOKS
1. Discrete Mathematics for computer scientists & Mathematicians, *J.L. Mott*, *A. Kandel*, *T.P.* Baker PHI 2. Discrete Mathematical Structures With Applications to Computer Science, JP Tremblay, R Manohar
REFERENCE BOOKS
1. Logic and Discrete Mathematics, *Grass Man & Trembley*, Pearson Education.

DATA STRUCTURES

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS05	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

1. Impart the basic concepts of structures, pointers and data structures.
2. Understand concepts linked lists and their applications.
3. Understand basic concepts about stacks, queues and their applications.
4. Understand basic concepts of trees, graphs and their applications.
5. Enable them to write algorithms for sorting and searching.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Use arrays, pointers and structures to formulate algorithms and programs.
2. Design and implement applications of Linked List.
3. Design and implement Stack ADT using Array and Linked List.
4. Design and implement Queue ADT using Array and Linked List.
5. Solve problems involving graphs and trees.
6. Analyze searching and sorting techniques based on time and space complexity.
7.

UNIT – I	INTRODUCTION TO DATA STRUCTURES	CLASSES: 15

Introduction to Structures: Structures definition, initialization ,accessing structures, nested structures, arrays of structures, structures and functions, self-referential structures ,basics of Pointers, dynamic memory allocation functions, pointers to structures.

Introduction to Data Structures: Introduction to Data Structures, abstract data types, Definition, Linear and Non-linear data structures, Array definition, initialization, 1-D Array,2-DArray and Multidimensional Arrays

UNIT - II	Linked List	CLASSES: 15

Singly Linked Lists-Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations-Insertion, Deletion, Doubly Linked Lists- Operations-Insertion, Deletion, Applications of Linked lists – Polynomials.

UNIT - III	STACKS	CLASSES: 12

Stacks-Stack ADT, definition, operations, array and linked implementations in C, **Applications-**infix to postfix conversion, Postfix expression evaluation, recursion implementation.

UNIT - IV	QUEUES	CLASSES: 12

Queues- Queue ADT, definition and operations ,array and linked Implementations in C, Circular queues-array and linked implementations in C, Dequeue (Double ended queue) ADT, array and linked implementations in C.

UNIT – V	SEARCHING & SORTING AND NON-LINEAR DATA STRUCTURES	CLASSES: 10

Searching- Linear Search, Binary Search, **Sorting**- Bubble Sort, Insertion Sort, Selection Sort, Quick sort, Merge Sort, Comparison of Sorting methods.
Non-Linear Data Structures- Trees – Introduction, Definition, Terminology, Applications, Tree Representations- List Representation, Left Child – Right Sibling Representation. **Graphs -** Introduction, Definition, Terminology, Applications.

TEXT BOOKS

1. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 6th Edition, 2012.
2. "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
3. Data Structures using C, R.Thareja 2nd Edition, Oxford Press.
4. Data Structures using C – A. S.Tanenbaum, Y. Langsam, and M.J. Augenstein, PHI/Pearson Education.

REFERENCE BOOKS

1. Data structures using C, A.K.Sharma, 2nd edition, Pearson.
2. "How to Solve it by Computer", 2nd Impression by R. G. Dromey, Pearson Education
3. Data Structures: A Pseudocode Approach with C, 2 nd Edition, R. F. Gilberg and B.A.Forouzan,

DATABASE MANAGENET SYSTEMS

II B.Tech -I Semester: IT/CSE/CSIT								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS09	PCC	3	-	-	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the concepts of database management systems and ER Diagrams.
2. Gain knowledge on relational model and integrity constraints.
3. Understand SQL queries and Normalization of relations.
4. Understand the transaction management and concurrency control techniques.
5. Understand the concepts of Triggers, Stored Procedures and DCL Commands

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Recognize the basic concepts and the applications of database systems.
2. Design ER-models to represent simple database application scenarios and convert ER-Model to Relational Model.
3. Demonstrate SQL queries and apply Normalization techniques.
4. Summarize the usage of different concurrency control protocols.
5. Demonstrate the role of DBA using DCL commands and Apply PL/SQL to interact with database. Also, to summarize the usage of tree-based indexing

UNIT-I	INTRODUCTION TO DATABASE AND DATABASE DESIGN	Classes: 11

INTRODUCTION: Introduction and applications of DBMS, Purpose of data base, Data Independence, Database System architecture- Levels, Database users and DBA.
DATABASE DESIGN: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model

UNIT-II	RELATIONAL MODEL , RELATIONAL ALGEBRA AND CALCULUS	Classes: 11

INTRODUCTION TO THE RELATIONAL MODEL: Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design: E-R to relational, introduction to views, destroying/altering tables and views
Relational Algebra, Relational Calculus: Tuple relational Calculus, Domain relational calculus

UNIT-III	INTRODUCTION TO SQL & SCHEMA REFINEMENT	Classes: 20

INTRODUCTION TO SQL: Basics of SQL:DDL, DML, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions – numeric, date, string functions, set operations, sub-queries, correlated sub-queries, Use of group by, having, order by, join and its types, Exist, Any, All
CONCEPT OF JOINS :Inner Join, Outer Join, Left Outer Join, Right Outer Join, Self-Join
SCHEMA REFINEMENT: Problems caused by redundancy, decompositions, problems related to decomposition, reasoning about functional dependencies, FIRST, SECOND, THIRD normal forms, BCNF, lossless join decomposition, multi-valued dependencies, FOURTH normal form, FIFTH normal form

UNIT-IV	TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL	Classes: 11
Introduction to Transactions: Transaction Concept, Transaction State, ACID Properties, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation- Based Protocols, Multiple Granularity, Recovery and Atomicity, Log–Based Recovery, Recovery with Concurrent Transactions. TCL Commands – Save point, Commit and Rollback		
UNIT-V	OVERVIEW OF STORAGE AND INDEXING. PL/SQL.	Classes: 11
Data on External Storage, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes. Tree based indexing: Intuitions for tree Indexes, Indexed Sequential Access Methods (ISAM), **B+ Trees:** A Dynamic Index Structure. **PL/SQL :** Overview of Triggers, Stored Procedures: triggers-Row level table level and active databases, Stored Procedures IN, OUT parameters. **DBA** – Introduction to DBA, Creating Users, Grant/Revoke Permissions on tables using DCL Commands		

Text Books:

2. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2011.
3. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill, 3rd Edition, 2007.
4. Michael McLaughlin, Oracle Database 11g PL/SQL Programming, Oracle press.
5. RamezElmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson / Addisionwesley, 2007

Reference Books:

1. Peter Rob, Carlos Coronel, Database Systems Design Implementation and Management, 7th edition, 2009.
2. Scott Urman, Michael McLaughlin, Ron Hardman, "Oracle database 10g PL/SQL programming ",6th edition, Tata McGraw Hill,2010
3. S.K.Singh, "Database Systems Concepts, Design and Applications", First edition, Pearson Education, 2006.
4. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013.

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

II B. TECH-ISEMESTER

Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6HS08	HSMC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSEOBJECTIVES

1. To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets,forms of business organizations, capital budgeting and financial accounting and financial analysis.

COURSEOUTCOMES

At the end of the course, the student will

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Develop an understanding of
4. Analyse how capital budgeting decisions are carried out.
5. Understanding the framework for both manual and computerized accounting process
6. Know how to analyse and interpret the financial statements through ratio analysis.

UNIT-I	Introduction & Demand Analysis	CLASSES: 10

Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

UNIT-II	Production & Cost Analysis: Production Function	CLASSES: 10

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cos Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance

UNIT-III	Markets & New Economic Environment	CLASSES: 12

Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of differer forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario

UNIT-IV	Capital Budgeting	CLASSES: 14

Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capita Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method Accounting Rate of return (ARR) and Net Present Value Method (simple problems).		
UNIT-V	**Introduction to Financial Accounting & Financial Analysis**	**CLASSES: 14**
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, and Trial Balance - Final Accounts (Trasing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		

TEXTBOOKS

1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009.
2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013.
3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.

REFERENCEBOOKS

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012.
3. Lipsey & Chrystel, Economics, Oxford University Press, 2012.
4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012.
5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.
6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja & Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

DATA STRUCTURES LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS06	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Ability to identify the appropriate data structure for given problem.
2. Effectively use compilers include library functions, debuggers and trouble shooting.
3. Write and execute programs using data structures such as arrays, linked lists to implement stacks,queues.
4. Write and execute programs in C to implement various sorting and searching.

COURSE OUTCOMES

The course should enable the students to:

1. Use appropriate data structure for given problem.
2. Use compilers include library functions, debuggers and trouble shooting.
3. Execute write programs in C to implement various types Linked Lists.
4. Execute programs using data structures such as arrays, linked lists to implement stacks.

LIST OF EXPERIMENTS

WEEK - 1 BASICS OF STRUTCURES

Write a C program using structures.

1. To read name, Roll number, year
2. To enter marks of three subjects of a student
3. To print the student the name, roll number, average marks of the student.

WEEK - 2 STRUTCURES Using FUNCTIONS and Pointers

Write a C Program using functions and pointers to

a. Reading a complex number
b. Writing a complex number
c. Add two complex numbers
d. Multiply two complex numbers

Note: represent structure using Pointers and Functions Concept .

WEEK - 3 ARRAYS

1. Write a C program

a) To add two matrices
b)To multiply two matrices

2. Write a C program to implement Sparse Matrices.

WEEK - 4 SINGLE LINKED LIST

Write a C program that uses functions to perform the following: a. Create TWO singly linked list of integers. b. Concatenate TWO Singly Linked Lists. c. Display the contents of the above list after concatenation.	
WEEK - 5	**DOUBLE LINKED LIST**
Write a C program that uses functions to perform the following: a. Create a doubly linked list of integers. b. Delete a given integer from the above doubly linked list. c. Display the contents of the above list after deletion.	
WEEK - 6	**STACK**
Write C programs to implement a STACK ADT using i)array ii) linked list	
WEEK - 7	**STACK APPLICATION**
1. Write a C program that uses stack operations to convert a given infix expression into its postfixEquivalent, Implement the stack using an array. **2. Write a C program that uses Stack to evaluate Postfix Expression.**	
WEEK - 8	**QUEUE**
Write C programs to implement a Queue ADT using i) array ii) linked list	
WEEK - 9	**DOUBLE ENDED QUEUE**
Write C programs to implement a double ended queue ADT using i) array ii) doubly linked list	
WEEK - 10	**SEARCHING**
Write C programs for implementing the following searching methods: a) Linear Search b) Binary Search	
WEEK - 11	**SORTING**
Write C programs for implementing the following sorting methods to arrange a list of integers inAscending order : a) Insertion sort b) Merge sort	
WEEK - 12	**SORTING**
Write C programs for implementing the following sorting methods to arrange a list of integers in ascending order: a) Quick sort b) Selection sort	
TEXT BOOKS	
1. C and Data Structures, Prof. P.S.Deshpande and Prof. O.G. Kakde, Dreamtech Press. 2. Data structures using C, A.K.Sharma, 2nd edition, Pearson. 3. Data Structures using C, R.Thareja, Oxford University Press.	

DATABASE MANAGEMENT SYSTEMS LAB

II B.Tech I Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credit	Maximum Marks		
A6CS10	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Use the basics of SQL and construct queries using SQL in database creation and interaction
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4. Analyze and Select storage and recovery techniques of database system.

COURSE OUTCOMES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Develop an ER model for a given database.
3. Use the basics of SQL and construct queries using SQL in database creation and interaction.
4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
5. Analyze and Select storage and recovery techniques of database system.
6. Develop Procedures, Cursors, and Triggers in database system.

LIST OF EXPERIMENTS

Week-1 DDL Commands

Creation of Tables using SQL- Overview of using SQL tool and Data types in SQL

a. Altering Tables and
b. Dropping Tables

Week-2 Create Table with Primary key and Foreign Key& DML Commands

Creating Tables (along with Primary and Foreign keys), Practicing DML commands-

a. Insert,
b. Update
c. Delete.

Week-3 Selection Queries

Practicing Select command using following operations

a. AND, OR
b. ORDER BY
c. BETWEEN
d. LIKE
e. Apply CHECK constraint

Week-4 AGGREGATE FUNCTIONS and Views

Practice Queries using following functions a. COUNT, b. SUM, c. AVG, d. MAX, e. MIN, Apply constraint on aggregation using a. GROUP BY, b. HAVING,	
Week-5	**Nested QUERIES**
Practicing Nested Queries using a. UNION, b. INTERSECT, c. CONSTRAINTS d. IN	
Week-6	**CO- RELATED NESTED QUERIES**
Practicing Co – Related Nested Queries using a. EXISTS, b. NOT EXISTS. ANY, ALL	
Week-7	**JOIN QUERIES**
Practicing Join Queries using a. Inner join b. Outer join c. Equi join d. Natural join	
Week-8	**TRIGGERS**
Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.	
Week-9	**PROCEDURES**
Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure	
Week-10	**CURSORS**
Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.	
Week-11	**PL/SQL Part 1**
Practice PL/SQL – a. block structure, b. variables, c. data types,	
Week-12	**PL/SQL Part 2**
Practice PL/SQL – a. operators, b. control structures;	

Case study 1: College Management
Case study 2 : An Enterprise/Organization
Case study 3 : Library Management system
Case study 4: Sailors and shipment system

Reference Books:

1. *Database System Concepts, by Silberschatz, Sudarshan, and Korth, 6th edition.*
2. Database management System by RaghuRamaKrishna, 3rd edition

SKILL DEVELOPMENT COURSE (HTML/CSS/JAVASCRIPT/BPPTSTRAP)

II B.Tech I Semester: IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6IT04	PCC	-	-	2	1	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To familiarize with the syntax and semantics of HTML
2. To learn the usage of CSS for styling web pages.
3. To use Javascript for creating dynamic and interactive web content like applications and browsers.
4. To use bootstrap for enabling responsive development of mobile-first websites
5. To develop a complete website by using HTML, CSS, JavaScript and bootstrap with some case studies.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Apply basic HTML for creating websites.
2. Apply CSS for styling web pages.
3. Demonstrate the usage of JavaScript for creating dynamic and interactive web content
4. Use Bootstrap for front-end development of creating websites and web apps.
5. Develop an interactive website using, HTML, CSS, JavaScript and bootstrap.

UNIT-I — Classes: 6

Introduction to HTML, Basic Formatting Tags, Grouping using DIV-Span, Headers, List, Images, Hyperlink, Table, iframes, miscellaneous Tags.

UNIT-II — Classes: 6

CSS Introduction, Types of CSS, Syntax-Selectors, Font, Box Model, Display Positioning, Background cursor, List Tables, Audio- video file creation

UNIT-III — Classes: 6

Introduction to Java Scripts, data types and variables, Operators, Statements, functions, Objects in Java Script, event handling, forms with Javascript, DOM, Dynamic HTML with Java Script.

UNIT-IV — Classes: 6

Introduction to Bootstrap, Syntax, Container, Bootstrap component, Advanced Component, Utilities

UNIT-V — Classes: 6

Case study 1: Create a Dynamic Web page for College Library Management System
Case Study 2: Create a website using animation

Text Books:

1. Web Programming, building internet applications, Chris Bates 2nd edition, WILEY Dreamtech (Unit 1,2,3,5)
2. Learning Bootstrap, Aravind Shenoy, Ulrich Sossou, December 2014, Packt Publishing, ISBN: 9781782161844 (Unit 4,5)

GENDER SENSITIZATION

II B. TECH-I SEMESTER

Course Code	Category	Hours /Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6HS05	MC	-	-	2	0	50	0	50
Contact Classes:16	Tutorial Classes:0	Practical Classes:00				Total Classes:16		

COURSEOBJECTIVES

1. To provide a critical perspective on the socialization of men and women.
2. To introduce students to information about some key biological aspects of genders.
3. To expose the students to debates on the politics and economics of work.
4. To help students reflect critically on gender violence.
5. To expose students to more egalitarian interactions between men and women.

COURSEOUTCOMES

1. Develop a better understanding of important issues related to gender in contemporary India.
2. Sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender
3. Attain a finer grasp of how gender discrimination works in our society and how to counter it
4. Men and Women students and professionals will be better equipped to work and live together as equals.
5. Create a sense of appreciation of women in all walks of life.

UNIT-I	UNDERSTANDING GENDER	CLASSES: 03

Introduction: Introduction to Gender, What is Gender, Why should we study it..
Socialization: Making Women, Making Men- Preparing for Woman hood. Growing up Male. First lessons in Caste: Different Masculinities.

UNIT-II	GENDER ROLES AND RELATIONS	CLASSES: 03

Two or Many? -Struggles with Discrimination-Missing Women-Sex Selection and Its Consequences Declining Sex Ratio. Demographic Consequences-Gender Spectrum: Beyond the Binary

UNIT-III	GENDER AND LABOUR	CLASSES: 03

Di Valuation of Labour-Housework: The Invisible Labor- “My Mother doesn’t Work.” “Share the Load.”-Work: Its Politics and Economics-Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.

UNIT-IV	GENDER-BASED VIOLENCE	CLASSES: 04

Sexual Harassment: Say No! -Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment-Further Reading: “Chupulu”. Domestic Violence: Speaking Out Is Home a Safe Place? -When Women Unite [Film].Rebuilding Lives.Thinking about Sexual Violence Blaming the Victim-“I Fought for my Life….”Additional Reading: The Caste Face of Violence.

UNIT-V	GENDERANDCOEXISTENCE	CLASSES: 03
Gender Issues- Just Relationships: Being Together as Equals Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers. Rosa Parks The Brave Heart..		
TEXT BOOKS		
All the five Units in the Textbook, "Towards a World of Equals: A Bilingual Textbook on Gender" written by A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad,Telangana State in the year 2015.		
REFERENCE BOOKS		
1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012 2. Abdulali Sohaila. "I Fought For My Life…and Won."Available online at: http://www.thealternative.in/lifestyle/i- 3. fought-for-my-life and-won-sohaila-abdulal/		

II B.TECH II SEMESTER
SYLLABUS

DIGITAL LOGIC DESIGN AND COMPUTER ORGANIZATION

<table>
<tr><td colspan="9">II B. TECH- II SEMESTER</td></tr>
<tr><td>Course Code</td><td>Category</td><td colspan="3">Hours / Week</td><td>Credits</td><td colspan="3">Maximum Marks</td></tr>
<tr><td rowspan="2">A6CS28</td><td rowspan="2">ESC</td><td>L</td><td>T</td><td>P</td><td>C</td><td>CIE</td><td>SEE</td><td>Total</td></tr>
<tr><td>3</td><td>-</td><td>-</td><td>3</td><td>40</td><td>60</td><td>100</td></tr>
</table>

COURSE OBJECTIVES

The course should enable the students to:

1. Understand different number systems and Boolean Algebra.
2. Design of combinational and sequential logic circuits
3. Understand different Computer Instructions
4. Understand concepts of register transfer logic and arithmetic operations.
5. Learn different types of memory hierarchy

COURSE OUTCOMES

1. Able to perform the conversion among different number systems.
2. Able to design combinational and sequential logic circuits
3. Able to understand different computer instructions
4. Identify basic components and design of the CPU
5. Compare various types of IO mapping and memory mapping techniques

UNIT - I	NUMBER THEORY and BOOLEAN ALGEBRA	CLASSES: 12

Representation of numbers of different radix, conversion of numbers from one radix to another radix, r-1‘s complement and r‘s complement. 4-bit codes. Basic Theorems and Properties of Boolean algebra, Canonical and Standard Forms, Digital Logic Gates, Universal Logic Gates. K-Map Method.

UNIT - II	COMBINATIONAL and SEQUENTIAL LOGIC CIRCUITS	CLASSES: 14

Design of Half adder, full adder, half subtractor, full subtractor. Decoder, Encoder, Multiplexer, De-multiplexer and comparator. basic flip-flops, truth tables and excitation tables (NAND RS latch, NOR RS latch, RS flip-flop, JK flip-flop, T flip-flop, D flip-flop with preset and clear terminals). Classification of sequential circuits (synchronous and asynchronous);

UNIT - III	DESIGN of COMPUTATIONAL LOGIC	CLASSES: 16

Conversion of flip-flops to other flip-flops. Design of Ripple counters, design of synchronous counters, Johnson counter, ring counter, shift register, bi-directional shift register, universal shift register.

Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control Memory-Reference Instructions, Input-Output and interrupt.

UNIT - IV	REGISTER TRANSFER AND MICRO-OPERATIONS:	CLASSES: 16

Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC),.Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit

UNIT - V	MEMORY SYSTEM	CLASSES: 16
INPUT OUTPUT ORGANIZATION: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA. **MEMORY ORGANIZATION**: Memory Hierarchy, Main memory, Auxiliary memory, Associate memory, Cache memory		
TEXT BOOKS		
1. Switching and Finite Automata Theory- ZviKohavi & Niraj K. Jha, 3rd Edition, Cambridge. 2. Digital Design- Morris Mano, PHI, 3rd Edition. 3. M. Morris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India. 4. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill		
REFERENCE BOOKS		
1. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd Ed,John Wiley &SonsInc. 2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersy. 3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson		

DESIGN AND ANALYSIS OF ALGORITHMS

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS15	PCC	3	-	-	3	40	60	100

COURSE OUTCOMES

At the end of this course students will be able to:

1. Identify various Time and Space complexities of various algorithms
2. Understand Tree Traversal method and Greedy Algorithms
3. Apply Dynamic Programming concept to solve various problems
4. Apply Backtracking, Branch and Bound concept to solve various problems
5. Implement different performance analysis methods for non deterministic algorithms

UNIT-I

INTRODUCTION: Algorithm, pseudo code for expressing algorithms, performance analysis-space complexity, time complexity, asymptotic notation- big (O) notation, omega notation, theta notation and little (o) notation, recurrences, probabilistic analysis, disjoint set operations, union and find algorithms.

UNIT-II

DIVIDE AND CONQUER: General method, applications-analysis of binary search, quick sort, merge sort, AND OR Graphs.

GREEDY METHOD: General method, Applications-job sequencing with deadlines, Fractional knapsack problem, minimum cost spanning trees, Single source shortest path problem.

UNIT-III

GRAPHS (Algorithm and Analysis): Breadth first search and traversal, Depth first search and traversal, Spanning trees, connected components and bi-connected components, Articulation points.

DYNAMIC PROGRAMMING: General method, applications - optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem, Reliability design.

UNIT-IV

BACKTRACKING: General method, Applications- n-queen problem, Sum of subsets problem, Graph coloring and Hamiltonian cycles.

BRANCH AND BOUND: General method, applications - travelling sales person problem, 0/1 knapsack problem- LC branch and bound solution, FIFO branch and bound solution.

UNIT-V

NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, non-deterministic algorithms, NP-hard and NP-complete classes.

TEXT BOOKS

1. Ellis Horowitz, Satraj Sahni, Rajasekharam (2007), Fundamentals of Computer Algorithms, 2nd edition, University Press, New Delhi.

REFERENCE BOOKS

1. R. C. T. Lee, S. S. Tseng, R.C. Chang and T. Tsai (2006), Introduction to Design and Analysis of Algorithms A strategic approach, McGraw Hill, India.
2. Allen Weiss (2009), Data structures and Algorithm Analysis in C++, 2nd edition, Pearson education, New Delhi.
3. Aho, Ullman, Hopcroft (2009), Design and Analysis of algorithms, 2nd edition, Pearson education, New Delhi

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

II B.Tech II Semester: IT/CSE/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6IT02	PCC	3	-	-	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Use object oriented programming concepts to solve real world problems.
2. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
3. Use multithreading concepts to develop inter process communication.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).
5. Solve real world problems using Collections

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Use object oriented programming concepts to solve real world problems.
2. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
3. Use multithreading concepts to develop inter process communication.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).
5. Build the internet-based dynamic applications using the concept of applets

UNIT-I	JAVA BASICS	Classes: 12

JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, type conversion and casting, simple java program, constructors, methods, Static block, Static Data, Static Method ,String and String Buffer Classes, Using Java API Document.

UNIT-II	INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES	Classes: 11

INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Method Overloading, Method overriding, Abstract classes, Encapsulation, Need for encapsulation in java, Data hiding vs Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword.
PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces

UNIT-III	EXCEPTION HANDLING AND FILES	Classes: 10

IEXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes.
I / O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, File Handling

UNIT-IV	MULTITHREADING AND JDBC	Classes: 10
MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication. **JDBC-Connecting to Database** - JDBC Type 1 to 4 drives, connecting to a database, querying a Database and processing the results, updating data with JDBC		
UNIT-V	**COLLECTION FRAMEWORK**	**Classes: 10**
COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, calender and Properties		

Text Books:

1. Herbert Schildt and Dale Skrien, ‖Java Fundamentals – A comprehensive Introduction‖, McGraw Hill, 1st Edition, 2013.
2. Herbert Schildt, ―Java the complete reference‖, McGraw Hill, Osborne, 7th Edition, 2011.
3. T.Budd, ―Understanding Object- Oriented Programming with Java‖, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.

Reference Books:

1. P.J.Dietel and H.M.Dietel , ―Java How to program‖, Prentice Hall, 6th Edition, 2005.
2. P.Radha Krishna , ―Object Oriented programming through Java‖, CRC Press, 1st Edition, 2007.
3. S.Malhotra and S. Choudhary, ―Programming in Java‖, Oxford University Press, 2nd Edition,

SOFTWARE ENGINEERING

II B. TECH- II SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6CS13	**PCC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	-	-	3	40	60	100

COURSE OBJECTIVES

1. Analyze software development life cycle and select appropriate model suited for diverse software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document
3. Conceptualize the system through design with emphases on architectural modeling and user interface
4. Classify software testing strategies and recommend testing techniques during the construction of software.
5. Examine the application of metrics and software tools during software development

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Understand software development life cycle and select appropriate model suited for diverse software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document
3. Conceptualize the system through design with emphases on architectural modeling and user
4. Classify software testing strategies and recommend testing techniques during the construction of software.
5. Examine the application of metrics and software tools during software development

UNIT - I	INTRODUCTION TO SOFTWARE ENGINEERING	CLASSES: 14

INTRODUCTION TO SOFTWARE ENGINEERING: The Evolving nature of software engineering, Changing nature of software engineering, Software engineering Layers, The Software Processes, Software Myths.

PROCESS MODELS: A Generic Process Model, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Spiral Model, The Unified Process, Personal and Team Process Models, the Capability Maturity Model Integration (CMMI).

UNIT - II	REQUIREMENTS ENGINEERING	CLASSES: 14

REQUIREMENTS ENGINEERING: Functional and Non-Functional Requirements, The Software requirements Document, Requirements Specification, requirements Engineering, Requirements Elicitation and Analysis, Requirement Validation, Requirement Management, System Modeling: Context Models, Interaction Models,

DESIGN CONCEPTS: The Design Process, Design Concepts, The Design Models, Architectural Design: Software Architecture, Architectural Genres, Architectural Styles.

UNIT - III	DESIGN IMPLEMENTATION	CLASSES: 14

DESIGN AND IMPLEMENTATION: Design Patterns, Implementation Issues, Open Source Development. User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, **DESIGN EVALUATION.**

SOFTWARE TESTING STRATEGIES: A Strategic approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Validation Testing, System Testing, The Art of Debugging, White-Box Testing, Black Box Testing. Structural Models, Behavioral Model, Model-Driven Engineering.

UNIT - IV	PRODUCT METRICS	CLASSES: 10

PRODUCT METRICS: A Frame Work for Product Metrics, Metrics for the Requirements Model, Metrics for Design Model, Metrics for Source Code, Metrics for Testing.

PROCESS AND PROJECT METRICES: Metrics in the Process and Project Domains, Software Measurements, Metrics for Software Quality, Risk Management: Risk verses Proactive Risk Strategies, Software Risks, Risk Identification, Risk Projection, Risk Refinements, Risk Mitigation Monitoring and Management (RMMM), The RMMM Plan.

UNIT - V	OVERVIEW OF QUALITY MANAGEMENT AND PROCESS IMPROVEMENT	CLASSES: 10

OVERVIEW OF QUALITY MANAGEMENT AND PROCESS IMPROVEMENT: Overview of SEI -CMM, ISO 9000, CMMI, PCMM, TQM and Six Sigma.

OVERVIEW OF CASE TOOLS: Software tools and environments: Programming environments; Project management tools; Requirements analysis and design modeling tools; testing tools; Configuration management tools;

TEXT BOOKS

1. Roger S. Pressman (2011), Software Engineering, A Practitioner's approach, 7th edition, McGraw Hill International Edition, New Delhi.
2. Sommerville (2001), Software Engineering, 9th edition, Pearson education, India.

REFERENCE BOOKS

1. K. K. Agarval, Yogesh Singh (2007), Software Engineering, 3rd edition, New Age International Publishers, India.
2. Lames F. Peters, Witold Pedrycz(2000), Software Engineering an Engineering approach, John Wiely & Sons, New Delhi, India.
3. Shely Cashman Rosenblatt (2006), Systems Analysis and Design, 6th edition, Thomson
4. Publications, India

OPERATING SYSTEMS

<table>
<tr><td colspan="10">II B. TECH- II SEMESTER</td></tr>
<tr><td rowspan="2">Course Code</td><td rowspan="2">Category</td><td colspan="3">Hours / Week</td><td>Credits</td><td colspan="3">Maximum Marks</td></tr>
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</table>

COURSE OBJECTIVES

1. To explain main components of OS and their working
2. To familiarize the operations performed by OS as a resource Manager
3. To familiarize with Dead Locks
4. To impart various scheduling policies of OS
5. To teach the different memory management techniques.

COURSE OUTCOMES

1. Analyze the different structures and services of operating system.
2. Analyze various algorithms used for OS services with respect to defined/chosen criteria.
3. Solve the resource allocation and sharing problems.
4. Assess different methods to solve OS problems.
6. Analyze the memory management approaches of operating systems.

UNIT-I	OPERATING SYSTEMS OVERVIEW	CLASSES: 14

OPERATING SYSTEMS OVERVIEW: Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems.
OPERATING SYSTEMS STRUCTURES: Operating system services and systems calls, system programs, operating system structure, operating systems generations

UNIT - II	PROCESS MANAGEMENT	CLASSES: 16

PROCESS MANAGEMENT: Process concepts, process state, process control block, scheduling queues, process scheduling, Scheduling algorithms, multithreaded programming.
CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson’s solution, synchronization hardware, semaphores, classic problems of synchronization, readers and writers problem, dining philosophers problem, monitors, synchronization examples(Solaris).

UNIT - III	DEAD LOCKS	CLASSES: 14

DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker’s algorithm.
MEMORY MANAGEMENT: Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing.

UNIT - IV	FILE SYSTEM	CLASSES: 12

FILE SYSTEM: Concept of a file, access methods, directory structure, file system mounting, file sharing, protection.
File system implementation: file system structure, file system implementation, directory

implementation, allocation methods, free-space management, efficiency and performance.

UNIT - V	MASS STORAGE STRUCTURE	CLASSES: 14

MASS STORAGE STRUCTURE: overview of mass storage structure, disk structure, disk attachment, disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage structure.

I/O: Hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance

TEXT BOOKS

1. Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006).
2. Operating System Principles, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006), 7th edition, Wiley India Private Limited, New Delhi.

REFERENCE BOOKS

1. Operating Systems, Internals and Design Principles, Stallings (2006), 5th edition, Pearson Education, India.
2. Modern Operating Systems, 2nd edition,Andrew S. Tanenbaum (2007), Prentice Hall of India, India
3. Operating systems, Deitel&Deitel (2008), 3rd edition, Pearson Education, India.

OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB

II B.Tech II Semester – IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6IT03	PCC	-	-	3	1.5	40	60	100

COURSE OUTCOMES:

1. Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
2. Understand the use of different exception handling mechanisms and concept of multithreading for robust and efficient application development.
3. Understand and implement concepts on file streams and operations in java programming for a a given application programs.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).

LIST OF EXPERIMENTS

WEEK-1 JAVA BASICS

a. Write a java program that prints all real solutions to the quadratic equation ax2+bx+c=0. Read in a, b, c and use the quadratic formula.
b. The Fibonacci sequence is defined by the following rule. The first two values in the sequence are 1 and 1. Every subsequent value is the sum of the two values preceding it. Write a java program that uses both recursive and non-recursive functions.

WEEK-2 ARRAYS

a. Write a java program to sort given list of integers in ascending order.
b. Write a java program to multiply two given matrices

WEEK-3 STRINGS

a. Write a java program to check whether a given string is palindrome.
b. Write a java program for sorting a given list of names in ascending order.

WEEK-4 OVERLOADING & OVERRIDING

a. Write a java program to implement method overloading and constructors overloading.
b. Write a java program to implement method overriding.

WEEK-5 INHERITANCE

a. Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.

WEEK-6 INTERFACES

a. Write a program to create interface A in this interface we have two method meth1 and meth2. Implements this interface in another class named MyClass.
b. Write a program to give example for multiple inheritance in Java.

WEEK-7 EXCEPTION HANDLING

a. Write a program that reads two numbers Num1 and Num2. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception Display the exception..

WEEK-8 I/O STREAMS

a. Write a java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes.

b. Write a java program that displays the number of characters, lines and words in a text file.

WEEK-9	**MULTI THREADING**
a. Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number	
WEEK-10	**GENERICS**
a. Write a Java program to swap two different types of data using Generics. b. Write a Java program to find maximum and minimum of two different types of data using Generics	
WEEK-11	**COLLECTIONS**
Create a linked list of elements. a. Delete a given element from the above list. b. Display the contents of the list after deletion	
WEEK-12	**CONNECTING TO DATABASE**
a. Write a java program that connects to a database using JDBC and does add, delete, modify and retrieve operations.	

TEXT BOOKS:

1. P. J. Deitel, H. M. Deitel, "Java for Programmers", Pearson Education, PHI, 4th Edition, 2007.
2. P. Radha Krishna, "Object Oriented Programming through Java", Universities Press, 2nd Edition, 2007
3. Bruce Eckel, "Thinking in Java", Pearson Education, 4th Edition, 2006.
4. Sachin Malhotra, Saurabh Chaudhary, "Programming in Java", Oxford University Press, 5th Edition, 2010.

OPERATING SYSTEMS LAB

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS12	PCC	-	-	3	1.5	40	60	100

COURSE OBJECTIVES

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. To provide an understanding of the design aspects of operating system concepts through simulation
5. Introduce basic Unix commands, system call interface for process management, inter process communication and I/O in Unix

COURSE OUTCOMES

After completion of the course Students is able to:

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Scripts to handle the File system in Linux.
3. Simulate and implement operating system concepts such as scheduling, deadlock management.
4. Implement operating system concepts such as file management and memory management.
5. Able to implement C programs using Unix system calls

LIST OF EXPERIMENTS

WEEK - 1 BASIC LINUX COMMANDS

1. Linux Commands & Introduction of Shell
2. Linux commands like man, passwd, tty, script, clear, date, cal, cp, mv, ln, rm, unlink, mkdir, rmdir, du, df, mount, umount, find, unmask, who, cat, tail, head, sort, nl, uniq, grep, cut, paste, join, comm, cmp , diff , tr, awk ,tar.

WEEK - 2 SHELL PROGRAMS

1. Write a shell program to swap two numbers using temporary variable
2. Write a shell program to check whether given number is palindrome or not
3. Write a shell program to check whether the given character is lower case, upper case, digit, special character

WEEK - 3 SHELL PROGRAMS

1. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.

2. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.	
WEEK - 4	**PROCESS SCHEDULING ALGORITHMS**
1. Write C programs to simulate the following CPU Scheduling algorithms a) FCFS b) SJF	
WEEK - 5	**PROCESS SCHEDULING ALGORITHMS**
1. Write C programs to simulate the following CPU Scheduling algorithms a) Round Robin b) priority	
WEEK - 6	**SYSTEM CALLS**
1. Write programs using the I/O system calls of UNIX/LINUX operating system (open, read, write, close, fcntl, seek, stat, opendir, readdir)	
WEEK – 7	**DEADLOCKS**
1. Write a C program to simulate Bankers Algorithm for Deadlock Avoidance. 2. Write a C program to simulate Bankers Algorithm for Deadlock Prevention.	
WEEK – 8	**SEMAPHORES**
1. Write a C program to implement the Producer – Consumer problem using semaphores using UNIX/LINUX system calls.	
WEEK – 9	**IPC MECHANISMS**
1. Write C programs to illustrate the following IPC mechanisms a) Pipes b) FIFOs	
WEEK - 10	**IPC MECHANISMS**
rite C programs to illustrate the following IPC mechanisms Message Queues b) Shared Memory	
WEEK - 11	**MEMORY MANAGEMENT TECHNIQUES**
1. Write C programs to simulate the following memory management techniques a) Paging b) Segmentation	
WEEK - 12	**PAGE REPLACEMENT MECHANISMS**
1. Write C programs to simulate Page replacement policies a) FCFS b) LRU c) Optimal	
TEXT BOOKS	
1. Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley	

2. Advanced programming in the Unix environment, W.R.Stevens, Pearson education.
3. Operating Systems – Internals and Design Principles, William Stallings, Fifth Edition–2005, Pearson Education/PHI
4. Operating System - A Design Approach-Crowley, TMH.
5. Modern Operating Systems, Andrew S Tanenbaum, 2nd edition, Pearson/PHI
6. UNIX Programming Environment, Kernighan and Pike, PHI/Pearson Education
7. UNIX Internals: The New Frontiers, U. Vahalia, Pearson Education

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CONSTITUTION OF INDIA

II B. TECH- II SEMESTER								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6HS06	**MC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	0	0	0	50	0	50

COURSE OBJECTIVES

Students will be able to:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

COURSE OUTCOMES

Students will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels.
5. Explain the functions and responsibilities of election commission of india and union public service commission.

UNIT - I HISTORY OF MAKING OF THE INDIAN CONSTITUTION

History of Making of the Indian Constitution: Introduction to the constitution of India, the making of the constitution and salient features of the constitution.

UNIT - II PHILOSOPHY OF THE INDIAN CONSTITUTION

Philosophy of the Indian Constitution: Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Rightto Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.

UNIT - III UNION GOVERNMENT

Union Government: Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court),

Jurisdiction of the supreme court.		
UNIT - IV	**STATE GOVERNMENT**	
State Government: State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)		
UNIT - V	**ELECTION COMMISSION**	
Election Commission: Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.		

TEXT BOOKS

1. M.V.Pylee,Indian Constitution Durga DasBasu,Human Rights in Constitutional Law,Prentice–HallofIndia Pvt. Ltd .New Delhi
2. Noorani,A.G.,(South Asia Human Rights Documentation Centre),Challenges to Civil Right), Callenges to Civil Rights Guarantees in India,Oxford University Press,2012
3. P.M.Bakshi, The constitution of India, Universal Law Publishing Co.,
4. The Constitution of India,1950 (Bare Act),Government Publication.
5. Dr.S.N.Busi, Dr. B.R.Ambedkar framing of Indian Constitution,1st Edition, 2015.

REFERENCE BOOKS

1. M.P.Jain,Indian ConstitutionLaw,7thEdn.,LexisNexis,2014.
2. D.D.Basu, Introduction to the Constitution of India,Lexis Nexis,2015.
3. Indian constitution at work, NCERT
4. Subash Kashyap, Indian Constitution, National Book Trust
5. J.A.Siwach, Dynamics of Indian Government & amp; Politics
6. D.C.Gupta, Indian Government and Politics
7. H.M.Sreevai,Constitutional Law of India,4theditionin3volumes(Universal Law Publication)
8. J.C.Johari, Indian Government and Politics Hans
9. J.RajIndian Government and Politics

PROFESSIONAL ELECTIVES

PROFESSIONAL ELECTIVE – I			PROFESSIONAL ELECTIVE - II		
S. No	Code	Subject	S. No	Code	Subject
1	A6AI02	Artificial Intelligence	1	A6CT05	Scripting Languages
2	A6IT23	Introduction to Data Science	2	A6AI11	Data Mining
3	A6IT24	Information security	3	A6IT26	Information Retrieval Systems
4	A6IT25	Mobile Application Development	4	A6IT28	Adhoc Sensor Networks
PROFESSIONAL ELECTIVE – III			PROFESSIONAL ELECTIVE - IV		
S. No	**Code**	**Subject**	**S. No**	**Code**	**Subject**
1	A6AI14	Natural Language Processing	1	A6CT06	Robotics Process Automation
2	A6IT29	Soft Computing	2	A6AI19	Pattern Recognition
3	A6DS02	Data Analytics Using R	3	A6IT33	E-Commerce
4	A6CY25	Block Chain Technology	4	A6DS21	Data Visualization
PROFESSIONAL ELECTIVE – V			**PROFESSIONAL ELECTIVE - VI**		
S. No	**Code**	**Subject**	**S. No**	**Code**	**Subject**
1	A6CS22	Distributed Computing	1	A6AI25	Quantum Computing
2	A6AI17	Deep Learning	2	A6IT31	High Performance Computing
3	A6CT07	UI/UX Design	3	A6AI12	Image Processing
4	A6AI22	Augmented Reality and Virtual Reality	4	A6DS14	Predictive Analytics

PROGRAM OUTCOMES (PO's):

Engineering Graduates will be able to:

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSO's):

PSO1: An ability to analyse a problem, design algorithm, identify and define the computing requirements appropriate to its solution and implement the same.

PSO2: An Ability to adapt to a rapidly changing environment by learning and employing new programming skills and technologies.

VISION STATEMENT

To create and nurture competent Engineers and managers who would be enterprise leaders in all parts of the world with aims of reaching the skies and touching the stars and yet feet firmly planted on the ground - good human beings steeped in ethical and moral values.

MISSION STATEMENT

MLR Institute of Technology is committed to providing a positive, professional and conducive learning environment where all students are inspired to achieve their potential and strive for excellence in a global society as dignified professionals with the cooperation of all stakeholders.

QUALITY POLICY

We, at MLRIT, are committed to Educate, Enrich and Excel, in imparting Professional Education, by top-quality faculty; who endeavor to mentor the students as turn-key solution providers, while striving continually to improve through team work, innovation and research.

GOALS OF MLRIT

Goals of Engineering education at undergraduate / graduate level:

- Equip students with industry - accepted career and life skills
- To create a knowledge warehouse for students
- To disseminate information on skills and competencies that are in use and in demand by the industry.
- To create learning environment where the campus culture acts as a catalyst to student fraternity to understand their core competencies, enhance their competencies and improve their career prospects.
- To provide base for lifelong learning and professional development in support of evolving career objectives, which include being informed, effective, and responsible participants within the engineering profession and in society.
- To prepare students for graduate study in Engineering and Technology.
- To prepare graduates to engineering practice by learning from professional engineering assignments.